



Quantum Entanglement

in Mesoscopic Systems

南京大学物理学院
School of Physics, Nanjing University

Quantum Transport Theory II

A chapter in the course-note series

张谷副教授课程讲义
Course instructor: Associate Professor Gu Zhang

纠缠部分由陈伟教授讲授
Entanglement lectures by Professor Wei Chen

Entanglement measures, Bell tests, mesoscopic entanglers, current-noise detection, pseudospin, beam-splitter statistics, and quantum erasure.

September 2026

Revised edition

About these notes

These notes develop the connection between a bipartite quantum state and the counts or currents used to probe it. Sections 1 and 2 establish the state and measurement language; Section 3 derives the conditions under which current noise reconstructs pair correlations. Sections 4–6 apply that language to pseudospins, exchange at beam splitters, and quantum erasure. The experimental drawings are idealized device schematics, not fabrication layouts or records of completed experiments.

This revision uses the supplied annotated *Entanglement .pdf* as its content source and the supplied English LaTeX bundle as its editorial base. Definitions and explicit calculations take precedence where those materials conflict. Source-page correspondence, substantial corrections, additions for teaching, and unresolved source details are documented separately in the Chinese revision report. These notes accompany Associate Professor Gu Zhang's course at the School of Physics, Nanjing University. The entanglement lectures were delivered by Professor Wei Chen.

How to use the calculations. Before applying an equation, identify the subsystem cut, the reference sea, the detector convention, and whether the statement concerns a conditional pair or the full emission ensemble. A normalized two-qubit state and a stationary current are different mathematical objects; the rate and counting window that connect them are made explicit in Section 3. Short checks and their answers appear beside the corresponding derivations.

Contents

About these notes **ii**

1 Bipartite entanglement and its measures **1**
1.1 What must be specified before calling a state entangled . . . 1
1.2 Schmidt decomposition from the coefficient matrix . . . 2
1.3 Reduced states, purity, and entropy . . . 3
1.4 Why reduced entropy is not a mixed-state measure . . . 4
1.5 Two-qubit concurrence . . . 5

2 Bell–CHSH tests: correlation, locality, and witness logic **5**
2.1 Conditioning is not signalling . . . 5
2.2 Local hidden variables and the CHSH bound . . . 6
2.3 The spin singlet violates CHSH . . . 7
2.4 Operational ingredients of a Bell test . . . 9

3 Mesoscopic spin entanglers and current-noise detection **10**
3.1 Operational entanglement resides in accessible modes . . . 10
3.2 From binary counts to a spin correlator . . . 11
3.3 Finite counting time and connected current noise . . . 13
3.4 Weak-tunnelling model for the emitted singlet . . . 15
3.5 Spin-resolved cross noise . . . 16

4 Orbital and Nambu pseudospin entanglement **18**
4.1 Orbital two-level systems . . . 18
4.1.1 Electron–hole orbital entanglement from a filled sea . . . 19
4.2 Nambu electron–hole pseudospin . . . 20
4.2.1 General Nambu analyzer direction . . . 22

5 Fermionic beam splitters: bunching and antibunching **24**
5.1 Spinless input: the determinant enforces antibunching . . . 24
5.2 A spin singlet bunches at an ideal beam splitter . . . 25
5.3 Electron–hole pairs invert the charge-noise intuition . . . 26
5.4 From a pair record to a noise measurement . . . 27

6 Quantum erasure by correlated detection **28**
6.1 Reference: a one-particle interferometer . . . 28
6.2 Which-way information removes the marginal fringe . . . 29
6.3 The eraser basis recovers conditional fringes . . . 30
6.4 When visibility measures concurrence . . . 31
6.5 Coincidences, connected noise, and no signalling . . . 32

A Notation and selected references **33**
A.1 Notation . . . 33

References **34**

1 Bipartite entanglement and its measures

Notation and conventions. An entanglement statement is meaningful only after declaring the physical bipartition, the available local operations, whether the state is pure or mixed, and the measure or witness being used. Throughout, $e > 0$ denotes the elementary charge magnitude, physical-spin Pauli matrices are $\boldsymbol{\sigma}$, orbital/lead pseudospin uses $\boldsymbol{\mu}$, and Nambu electron–hole pseudospin uses $\boldsymbol{\tau}$. Unit analyzer directions are bold italic, for example $\mathbf{a}$, and obey $\|\mathbf{a}\| = 1$; counting and current operators are bold upright. This convention keeps directions and operators distinct without hat accents. Unless stated otherwise, logarithms are natural and tensor-product factors are ordered from left to right as they appear in the corresponding ket.

1.1 What must be specified before calling a state entangled

Quantum superposition, interference, and tunnelling already distinguish a single quantum degree of freedom from a classical particle. Entanglement is a property of a *composite* system: its state may fail to admit a mixture of independently prepared subsystem states. The Einstein–Podolsky–Rosen discussion provides a historical starting point for studying such correlations [1]. Entanglement and Bell nonlocality are related but distinct notions; section 2 specifies their relationship.

Let the physical bipartition be

$$\mathrm{H}_{AB} = \mathrm{H}_A \otimes \mathrm{H}_B, \qquad \rho_A = \mathrm{Tr}_B\, \rho_{AB}, \qquad \rho_B = \mathrm{Tr}_A\, \rho_{AB}. \tag{1.1}$$

The labels A and B must denote physical subsystems, modes, regions, leads, or local observable algebras. They are not automatically the formal labels of two identical particles.

Definition 1.1 (Product and entangled pure states)**.** A normalized pure state is a product state when there are normalized local vectors for which

$$|\Psi_{\mathrm{prod}}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle. \tag{1.2}$$

It is entangled relative to the declared bipartition when no such factorization exists.

For a product state, a local operation acts on only one tensor factor. If M_m is the measurement operator for outcome m on A, then

$$(M_m \otimes \mathbb{1}_B)|\Psi_{\mathrm{prod}}\rangle = (M_m|\psi_A\rangle) \otimes |\psi_B\rangle. \tag{1.3}$$

After normalization, the state of B is still $|\psi_B\rangle$ for every outcome with nonzero probability. A coherent superposition of product terms need not factor in this way. For four normalized local kets, consider

$$\begin{aligned} |\Psi\rangle &= \frac{1}{N}\left(|\psi_A\rangle|\psi_B\rangle + |\phi_A\rangle|\phi_B\rangle\right), \\ N^2 &= 2 + 2\,\mathrm{Re}[\langle\psi_A \mid \phi_A\rangle\langle\psi_B \mid \phi_B\rangle]\,. \end{aligned} \tag{1.4}$$

Here $N > 0$ is required; exact cancellation does not define a state, and $|\psi_X\rangle \neq |\phi_X\rangle$ for $X = A, B$. Separability is decided by the linear dependence of the two local pairs, not merely by the number of written terms. The prefactor reduces to $1/\sqrt{2}$ when the two product terms are orthogonal, for example when either local overlap vanishes.

Projecting A onto a normalized $|m\rangle$ leaves the unnormalized conditional state

$$|\widetilde{\psi}_{B|m}\rangle = \frac{1}{N}\left(\langle m \mid \psi_A\rangle|\psi_B\rangle + \langle m \mid \phi_A\rangle|\phi_B\rangle\right). \tag{1.5}$$

Its squared norm is the outcome probability p_m; for $p_m > 0$ the normalized state is $|\psi_{B|m}\rangle = |\widetilde{\psi}_{B|m}\rangle/\sqrt{p_m}$. Its relative coefficients generally depend on m. This is the local measurement

meaning of the correlation; the no-signalling statement about unconditioned statistics is treated in section 2.

A state can have many different expansions in product bases, so failure to guess a one-term expansion is not a separability test. Schmidt decomposition provides the basis-independent criterion developed next.

1.2 Schmidt decomposition from the coefficient matrix

For finite-dimensional spaces of dimensions d_A, d_B, choose orthonormal local bases $\{|i_A\rangle\}$ and $\{|j_B\rangle\}$. Every pure state has a coefficient matrix D,

$$|\Psi\rangle = \sum_{i,j} D_{ij}|i_A\rangle|j_B\rangle, \qquad D_{ij} = \langle i_A|\langle j_B|\Psi\rangle. \tag{1.6}$$

If the local basis columns transform by unitaries U_A and U_B,

$$|\tilde{i}_A\rangle = \sum_p (U_A)_{pi}|p_A\rangle, \qquad |\tilde{j}_B\rangle = \sum_q (U_B)_{qj}|q_B\rangle, \tag{1.7}$$

then the coefficient matrix in the new basis transforms as

$$\begin{aligned} \widetilde{D}_{ij} &= \langle\tilde{i}_A|\langle\tilde{j}_B|\Psi\rangle, &(1.8)\\ &= \sum_{p,q} (U_A^\dagger)_{ip}(U_B^\dagger)_{jq}\langle p_A|\langle q_B|\Psi\rangle, &(1.9)\\ &= \sum_{p,q} (U_A^\dagger)_{ip} D_{pq} (U_B^*)_{qj} = \left(U_A^\dagger D U_B^*\right)_{ij}. &(1.10) \end{aligned}$$

The complex conjugation on the right factor follows from treating the two ket bases independently.

Theorem 1.2 (Schmidt decomposition)**.** *Let $D = U\Sigma V^\dagger$ be a singular-value decomposition, where the nonzero diagonal entries s_k of Σ are real and positive. Define*

$$|k_A^{\mathrm{S}}\rangle = \sum_i U_{ik}|i_A\rangle, \qquad |k_B^{\mathrm{S}}\rangle = \sum_j V_{jk}^*|j_B\rangle. \tag{1.11}$$

Then

$$|\Psi\rangle = \sum_{k=1}^{r_{\mathrm{S}}} s_k |k_A^{\mathrm{S}}\rangle|k_B^{\mathrm{S}}\rangle = \sum_{k=1}^{r_{\mathrm{S}}} \sqrt{\lambda_k}|k_A^{\mathrm{S}}\rangle|k_B^{\mathrm{S}}\rangle, \qquad \lambda_k = s_k^2, \qquad \sum_k \lambda_k = 1. \tag{1.12}$$

The state is separable if and only if the Schmidt rank r_{S} is one.

The decomposition also uses the fewest possible product terms. Each product ket contributes a rank-one coefficient matrix, so an expansion with q nonzero product terms has rank $D \le q$. The singular-value decomposition constructs an expansion with exactly $r_{\mathrm{S}} = \operatorname{rank} D$ terms. Hence r_{S} is not merely the number of terms displayed in one convenient basis: it is the minimum number required by any product-state expansion.

Schmidt-rank classification for a bipartite pure state

$$\begin{cases} r_{\mathrm{S}} = 1 & \Longleftrightarrow \quad \text{separable,} \\ r_{\mathrm{S}} > 1 & \Longleftrightarrow \quad \text{entangled.} \end{cases}$$

Figure 1.1: Schmidt-rank classification of bipartite pure states. Rank one corresponds to a product state, whereas rank greater than one signals entanglement.

The singular values s_k and Schmidt probabilities λ_k must not be interchanged. The multiset of nonzero λ_k is unique. The Schmidt vectors may change by opposite local phases and, for degenerate λ_k, by a unitary rotation W on one side paired with W^* on the other. Relabelling terms and choosing arbitrary bases in unused zero-eigenvalue subspaces also change the written decomposition without changing the state. A passive change of basis, as in equation (1.7), changes the coordinates of the same state; an active local unitary changes the state while preserving its Schmidt spectrum.

1.3 Reduced states, purity, and entropy

Starting from equation (1.12), the reduced state follows without guessing a basis:

$$\begin{aligned}\rho_A &= \mathrm{Tr}_B|\Psi\rangle\langle\Psi| \\ &= \mathrm{Tr}_B \sum_{i,j} \sqrt{\lambda_i\lambda_j}|i_A^{\mathrm{S}}\rangle\langle j_A^{\mathrm{S}}| \otimes |i_B^{\mathrm{S}}\rangle\langle j_B^{\mathrm{S}}| \\ &= \sum_i \lambda_i |i_A^{\mathrm{S}}\rangle\langle i_A^{\mathrm{S}}|. \end{aligned} \tag{1.13}$$

Similarly,

$$\rho_B = \sum_i \lambda_i |i_B^{\mathrm{S}}\rangle\langle i_B^{\mathrm{S}}|. \tag{1.14}$$

Thus the two reduced states have the same nonzero spectrum. In the original coefficient bases one can check the conjugations directly:

$$\rho_A = DD^\dagger, \qquad \rho_B = D^{\mathsf{T}}D^*. \tag{1.15}$$

For example, $(\rho_B)_{jk} = \sum_i D_{ij}D^*_{ik}$; this is why the second Schmidt basis contains the columns of V^* rather than V.

Define the reduced purity and von Neumann entropy by

$$\gamma_A \equiv \mathrm{Tr}\,\rho_A^2 = \sum_i \lambda_i^2, \qquad S_A \equiv -\,\mathrm{Tr}(\rho_A \ln \rho_A) = -\sum_i \lambda_i \ln \lambda_i. \tag{1.16}$$

For a globally pure state,

$$\gamma_A = 1 \iff r_{\mathrm{S}} = 1 \iff |\Psi\rangle \text{ is separable}, \qquad \gamma_A < 1 \iff |\Psi\rangle \text{ is entangled}. \tag{1.17}$$

This criterion is restricted to globally pure states. With $d_{\min} = \min(d_A, d_B)$, normalization further gives $1/d_{\min} \le \gamma_A \le 1$ and $0 \le S_A \le \ln d_{\min}$. The entangled pure state is maximal across this cut when all $d_{\min}$ Schmidt probabilities are equal. We use $0 \ln 0 = 0$ by continuity.

For the spin singlet

$$|\Psi^-\rangle = \frac{|{\uparrow_A}{\downarrow_B}\rangle - |{\downarrow_A}{\uparrow_B}\rangle}{\sqrt{2}}, \tag{1.18}$$

the Schmidt probabilities are $1/2, 1/2$. Hence

$$S_A = -2\left(\frac{1}{2}\log_2\frac{1}{2}\right) = 1 \text{ bit}, \tag{1.19}$$

or $S_A = \ln 2$ nats when entropy is defined with the natural logarithm.

Under a local unitary $U_A \otimes U_B$,

$$\begin{aligned}\rho'_A &= \mathrm{Tr}_B\big[(U_A \otimes U_B)|\Psi\rangle\langle\Psi|(U_A^\dagger \otimes U_B^\dagger)\big] \\ &= U_A \rho_A U_A^\dagger. \end{aligned} \tag{1.20}$$

The eigenvalues λ_i, and therefore γ_A and S_A, do not change. This invariance is a basic requirement for a pure-state entanglement measure. A joint unitary that cannot be factorized as $U_A \otimes U_B$ need not preserve the Schmidt spectrum and can therefore generate or remove entanglement across the declared $A|B$ cut.

Tracing out B removes access to correlations with B. For a globally pure state, the resulting entropy measures precisely this entanglement-induced loss of locally accessible information. It is not ordinary thermodynamic ignorance about a mixed global preparation.

1.4 Why reduced entropy is not a mixed-state measure

A mixed state is *separable* when it has a convex product decomposition,

$$\rho_{AB}^{\text{sep}} = \sum_k p_k \, \rho_A^{(k)} \otimes \rho_B^{(k)}, \qquad p_k \geq 0, \quad \sum_k p_k = 1. \tag{1.21}$$

Each $\rho_X^{(k)}$ is a normalized positive density operator. Local states in different terms need not be orthogonal or commute. If no such decomposition exists, the mixed state is entangled.

Consider the more restrictive, classically correlated example

$$\rho_{AB}^{\text{cc}} = \sum_i p_i \, |i_A\rangle\langle i_A| \otimes |i_B\rangle\langle i_B|, \qquad p_i \geq 0, \qquad \sum_i p_i = 1. \tag{1.22}$$

Its reduced state is $\rho_A = \sum_i p_i |i_A\rangle\langle i_A|$ and can have positive entropy, even though equation (1.22) is an explicit convex mixture of product states. Therefore $S(\rho_A)$ alone is not an entanglement measure for a mixed ρ_{AB}.

More generally, a *classical–classical* state is diagonal in some orthonormal product basis,

$$\rho_{AB}^{\text{CC}} = \sum_{i,j} p_{ij} |i_A\rangle\langle i_A| \otimes |j_B\rangle\langle j_B|, \qquad p_{ij} \geq 0, \quad \sum_{i,j} p_{ij} = 1. \tag{1.23}$$

Such states form a subset of separable states. General separable mixtures can still have quantum discord. Zero discord on side A means that some complete rank-one projective measurement on A leaves the joint state unchanged when the outcome is ignored. Discord therefore depends on which side is measured; it is not an entanglement measure [5]. Thus "not entangled" does not by itself mean "classical in every sense."

For a concrete comparison, define the incoherent spin mixture

$$\rho_{\text{anti}} = \tfrac{1}{2}(|{\uparrow\downarrow}\rangle\langle{\uparrow\downarrow}| + |{\downarrow\uparrow}\rangle\langle{\downarrow\uparrow}|). \tag{1.24}$$

It and the singlet have identical reduced states $\mathbb{1}/2$ and identical anticorrelation along z, but only the singlet retains the off-diagonal coherences between the two product kets.

Joint state	$\langle \sigma_z \otimes \sigma_z \rangle$	$\langle \sigma_x \otimes \sigma_x \rangle$	ρ_A
Incoherent mixture ρ_{anti}	-1	0	$\mathbb{1}/2$
Singlet $\lvert\Psi^-\rangle\langle\Psi^-\rvert$	-1	-1	$\mathbb{1}/2$

Figure 1.2: Changing the measured spin component reveals different joint correlations for two fixed states. It does not turn a separable state into an entangled one. Both states have maximally mixed local states, so neither a single anticorrelated basis nor local entropy alone distinguishes them.

The quantum mutual information $I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ measures total correlations. It is $\ln 2$ for ρ_{anti} and $2 \ln 2$ for the singlet. Correlated outcomes, nonzero total correlation, quantum discord, entanglement, and Bell nonlocality are therefore not interchangeable criteria.

1.5 Two-qubit concurrence

For a normalized pure state of two qubits, with $\mathrm{H}_A = \mathrm{H}_B = \mathbb{C}^2$, concurrence is [4]

$$C(|\Psi\rangle) = \sqrt{2\,(1 - \operatorname{Tr}\rho_A^2)}, \qquad 0 \le C \le 1. \tag{1.25}$$

Because a two-qubit Schmidt spectrum has two elements (including a zero when the state is a product),

$$C = 2\sqrt{\lambda_1\lambda_2} = \left|\langle\Psi|(\sigma_y^{(A)} \otimes \sigma_y^{(B)})|\Psi^*\rangle\right|, \tag{1.26}$$

where $\sigma_y^{(A)}$ and $\sigma_y^{(B)}$ are Pauli-y matrices in the declared computational bases and $|\Psi^*\rangle$ is the corresponding complex conjugate state. These matrices refer to the mathematical qubits; when the qubit is orbital or Nambu rather than real spin, the spin flip is a mathematical construction, not an instruction to reverse a physical spin.

For $|\Psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$, normalized so that $|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$, the coefficient matrix is $D = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Using $\det\rho_A = |\det D|^2 = \lambda_1\lambda_2$ gives the useful check

$$C = 2|ad - bc|. \tag{1.27}$$

For a mixed two-qubit state $\rho = \rho_{AB}$, the appropriate extension is the minimum average pure-state concurrence over all ensemble decompositions,

$$C(\rho) = \min_{\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|} \sum_k p_k C(|\psi_k\rangle). \tag{1.28}$$

Wootters gives an explicit evaluation [4]. In the ordered product basis $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$, set

$$\tilde\rho = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y), \qquad R = \sqrt{\sqrt{\rho}\,\tilde\rho\,\sqrt{\rho}}. \tag{1.29}$$

Here the star means entrywise complex conjugation in that basis, and all square roots are the positive operator square roots. If $r_1 \ge r_2 \ge r_3 \ge r_4 \ge 0$ are the eigenvalues of the Hermitian positive matrix R, then

$$C(\rho) = \max(0, r_1 - r_2 - r_3 - r_4). \tag{1.30}$$

Equivalently, r_i are square roots of the eigenvalues of $\rho\tilde\rho$; that product is generally not Hermitian. This avoids confusing its eigenvalues with the Schmidt probabilities.

The unqualified symbol C in these notes is restricted to two-qubit concurrence. In larger pure bipartite systems the expression $\sqrt{2(1 - \operatorname{Tr}\rho_A^2)}$ is a useful generalized quantity, but its maximum is $\sqrt{2(1 - 1/d_{\min})}$, not generally one; the two-qubit spin-flip formula does not extend unchanged. The pure-state identity in equation (1.25) does not apply directly to a mixed ρ_{AB}.

Exercise 1.3 (A pure-state check, with answer)**.** For $|\psi_\eta\rangle = \sqrt{\eta}|00\rangle + \mathrm{e}^{\mathrm{i}\phi}\sqrt{1-\eta}|11\rangle$, $0 \le \eta \le 1$, find the reduced purity, entropy, and concurrence. Does the phase change the entanglement?

Answer. The Schmidt probabilities are $\eta, 1-\eta$, so $\gamma_A = \eta^2 + (1-\eta)^2$, $S_A = -\eta\ln\eta - (1-\eta)\ln(1-\eta)$, and $C = 2\sqrt{\eta(1-\eta)}$. The phase is removable by a local unitary. At $\eta = 0, 1$ the state is a product; at $\eta = 1/2$, $C = 1$ and $S_A = \ln 2$. In contrast, ρ_{anti} has $\gamma_A = 1/2$ but $C = 0$ because it is an explicit separable mixture.

2 Bell–CHSH tests: correlation, locality, and witness logic

2.1 Conditioning is not signalling

For the singlet equation (1.18), if Alice measures spin along the z axis and obtains $\uparrow$, Bob's conditional state is $\downarrow$ along the same axis, and conversely. This conditional update is independent of the spatial separation, which motivated the historical concern about an "instantaneous" collapse.

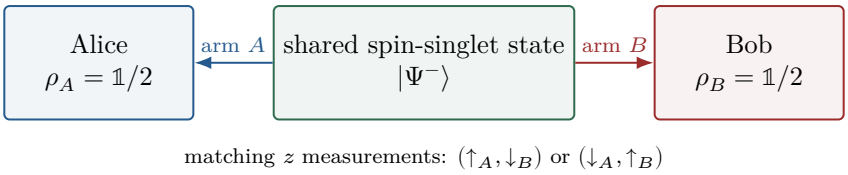


Figure 2.1: An Alice–Bob singlet pair and its maximally mixed local states. The outward arrows denote separated spatial arms. The lower row lists the anticorrelated outcome pairs for matching z measurements, not a pre-existing assignment of local spins or a signalling channel.

No controllable signal is transmitted by this update. Before Alice communicates her setting and result, Bob assigns

$$\rho_B = \mathrm{Tr}_A |\Psi^-\rangle\langle\Psi^-| = \frac{\mathbb{1}}{2}, \tag{2.1}$$

so his local statistics do not reveal what Alice measured. Comparing the correlated records requires ordinary information transfer and therefore cannot exceed the speed of light.

This holds for any joint state, including a mixed one. Let Alice's setting be x and let $M_{m\nu}^{(x)}$ be the Kraus operators of her local instrument; m is the recorded outcome and ν labels any unresolved alternatives. They act only on H_A and satisfy

$$\sum_{m,\nu} M_{m\nu}^{(x)\dagger} M_{m\nu}^{(x)} = \mathbb{1}_A. \tag{2.2}$$

The probability and conditional state are determined together by

$$p(m|x)\rho_{B|m,x} = \mathrm{Tr}_A \sum_{\nu} (M_{m\nu}^{(x)} \otimes \mathbb{1}_B)\rho_{AB}(M_{m\nu}^{(x)\dagger} \otimes \mathbb{1}_B). \tag{2.3}$$

Taking the full trace on the right gives $p(m|x)$; division is meaningful only when it is nonzero. Ignoring Alice's outcome gives

$$\begin{aligned} \rho_B^{(x)} &= \sum_m p(m|x)\rho_{B|m,x} \\ &= \mathrm{Tr}_A \left[\left(\sum_{m,\nu} M_{m\nu}^{(x)\dagger} M_{m\nu}^{(x)} \otimes \mathbb{1}_B \right) \rho_{AB} \right] = \rho_B. \end{aligned} \tag{2.4}$$

The middle step uses cyclicity of the partial trace for operators acting on A. Selecting one outcome can change Bob's conditional ensemble; summing over a complete local operation cannot change any Bob-only probability.

The singlet is invariant under a common local basis rotation. For $U \in \mathrm{SU}(2)$,

$$(U \otimes \mathbb{1})|\Psi^-\rangle = (\mathbb{1} \otimes U^\dagger)|\Psi^-\rangle, \tag{2.5}$$

which follows from $(U \otimes U)|\Psi^-\rangle = |\Psi^-\rangle$. This identity relates two expressions for the same joint state; neither side gives Alice control over Bob's local statistics.

2.2 Local hidden variables and the CHSH bound

A local hidden-variable model uses a variable λ with normalized density $\varrho(\lambda)$. Alice chooses $\mathbf{a}$ or $\mathbf{a}'$ and Bob chooses $\mathbf{b}$ or $\mathbf{b}'$. A deterministic local model assigns binary outcomes

$$A(\mathbf{a},\lambda) = \pm 1, \qquad B(\mathbf{b},\lambda) = \pm 1, \qquad \int \varrho(\lambda)\,\mathrm{d}\lambda = 1. \tag{2.6}$$

Stochastic local models can be absorbed into a larger λ, so the binary form is sufficient for the inequality. Locality means that Alice's response $A(\mathbf{a}, \lambda)$ contains no dependence on Bob's selected setting, and Bob's response $B(\mathbf{b}, \lambda)$ contains no dependence on Alice's selected setting. Measurement-setting independence further requires the distribution of λ to be independent of which pair of settings is selected,

$$\varrho(\lambda \mid \mathbf{a}, \mathbf{b}) = \varrho(\lambda). \tag{2.7}$$

More explicitly, a stochastic local model factorizes the response probabilities as

$$P(\alpha, \beta|\mathbf{a}, \mathbf{b}, \lambda) = P_A(\alpha|\mathbf{a}, \lambda)P_B(\beta|\mathbf{b}, \lambda), \quad \alpha, \beta = \pm 1. \tag{2.8}$$

The shared variable can carry arbitrary correlations from the preparation. Bell locality imposes this factorization, not statistical independence of the observed outcomes. The CHSH construction makes these assumptions experimentally testable [2, 3].

The correlator is

$$E(\mathbf{a}, \mathbf{b}) = \int \mathrm{d}\lambda\, \varrho(\lambda) A(\mathbf{a}, \lambda) B(\mathbf{b}, \lambda). \tag{2.9}$$

The symbol E emphasizes that this quantity is a signed correlator rather than a probability.

At fixed λ, use the CHSH sign convention

$$\begin{aligned} \beta(\lambda) = {} & A(\mathbf{a}, \lambda)B(\mathbf{b}, \lambda) - A(\mathbf{a}, \lambda)B(\mathbf{b}', \lambda) \\ & + A(\mathbf{a}', \lambda)B(\mathbf{b}, \lambda) + A(\mathbf{a}', \lambda)B(\mathbf{b}', \lambda). \end{aligned} \tag{2.10}$$

Factor the two Alice outcomes:

$$\begin{aligned} \beta(\lambda) = {} & A(\mathbf{a}, \lambda)\big[B(\mathbf{b}, \lambda) - B(\mathbf{b}', \lambda)\big] \\ & + A(\mathbf{a}', \lambda)\big[B(\mathbf{b}, \lambda) + B(\mathbf{b}', \lambda)\big]. \end{aligned} \tag{2.11}$$

Since $B, B' = \pm 1$, exactly one bracket in equation (2.11) vanishes and the other equals ± 2. Therefore

$$|\beta(\lambda)| = 2. \tag{2.12}$$

Averaging and using the triangle inequality gives

$$B_{\mathrm{CHSH}} = E(\mathbf{a}, \mathbf{b}) - E(\mathbf{a}, \mathbf{b}') + E(\mathbf{a}', \mathbf{b}) + E(\mathbf{a}', \mathbf{b}'), \qquad |B_{\mathrm{CHSH}}| \le 2. \tag{2.13}$$

Thus the bound uses outcomes bounded in $[-1, 1]$ (binary here), local response functions, and a setting-independent hidden-variable distribution. The same bound holds for local averages in $[-1, 1]$, since $|b - b'| + |b + b'| = 2\max(|b|, |b'|) \le 2$.

Remark 2.1 (The three-direction special case). The symmetric CHSH bound $|E(\mathbf{a}, \mathbf{b}) - E(\mathbf{a}, \mathbf{b}')| + |E(\mathbf{a}', \mathbf{b}) + E(\mathbf{a}', \mathbf{b}')| \le 2$ also follows from the same local model. If one additionally imposes perfect same-axis anticorrelation $E(\mathbf{b}', \mathbf{b}') = -1$ and sets $\mathbf{a}' = \mathbf{b}'$, it reduces to

$$|E(\mathbf{a}, \mathbf{b}) - E(\mathbf{a}, \mathbf{b}')| \le 1 + E(\mathbf{b}', \mathbf{b}). \tag{2.14}$$

This is the three-direction Bell form associated with the ideal singlet discussion [2]. CHSH does not require the extra perfect anticorrelation assumption.

2.3 The spin singlet violates CHSH

For physical-spin Pauli matrices $\boldsymbol{\sigma}$, the singlet predicts

$$E(\mathbf{a}, \mathbf{b}) = \langle \Psi^- \mid (\mathbf{a}\cdot\boldsymbol{\sigma}) \otimes (\mathbf{b}\cdot\boldsymbol{\sigma}) \mid \Psi^- \rangle = -\mathbf{a}\cdot\mathbf{b}. \tag{2.15}$$

Exercise 2.2 (Singlet correlation, with answer)**.** Derive equation (2.15). One route is to use $\langle \Psi^- \mid \sigma_i \otimes \sigma_j \mid \Psi^- \rangle = -\delta_{ij}$.

Answer. Acting with $\sigma_z \otimes \sigma_z$ on each singlet term gives -1, and $\sigma_x \otimes \sigma_x$ or $\sigma_y \otimes \sigma_y$ interchanges the two terms, again giving -1. The off-diagonal tensor components vanish. Expanding the analyzer product yields $\sum_{i,j} a_i b_j (-\delta_{ij}) = -\mathbf{a} \cdot \mathbf{b}$.

Choose four coplanar analyzer directions with polar angles

$$\theta_a = 0, \qquad \theta_b = \frac{\pi}{4}, \qquad \theta_{a'} = \frac{\pi}{2}, \qquad \theta_{b'} = \frac{3\pi}{4}. \tag{2.16}$$

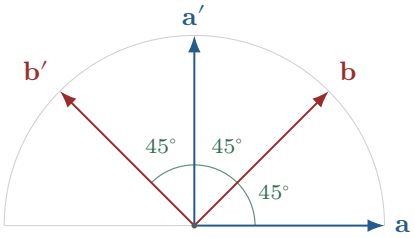


Figure 2.2: Coplanar analyzer directions that yield the maximal singlet violation for the sign convention in equation (2.13). Blue denotes Alice's two choices and red Bob's; adjacent rays are separated by 45°. All four rays represent unit directions.

Equation (2.15) gives

$$E(\mathbf{a}, \mathbf{b}) = -\frac{\sqrt{2}}{2}, \qquad E(\mathbf{a}, \mathbf{b}') = +\frac{\sqrt{2}}{2}, \tag{2.17}$$

$$E(\mathbf{a}', \mathbf{b}) = -\frac{\sqrt{2}}{2}, \qquad E(\mathbf{a}', \mathbf{b}') = -\frac{\sqrt{2}}{2}. \tag{2.18}$$

Thus

$$|B_{\mathrm{CHSH}}| = 2\sqrt{2} > 2, \tag{2.19}$$

which saturates the quantum Tsirelson bound. For completeness, write the CHSH operator as $\mathcal{B} = A \otimes (B - B') + A' \otimes (B + B')$, where the four Hermitian observables square to the identity. Direct multiplication gives

$$\mathcal{B}^2 = 4\mathbb{1} + [A, A'] \otimes [B, B'], \qquad \|\mathcal{B}\| \le 2\sqrt{2}, \tag{2.20}$$

because each commutator has operator norm at most two. The plus sign is tied to the minus sign on $E(\mathbf{a}, \mathbf{b}')$ in our CHSH convention.

Remark 2.3 (Witness, not measure). A valid CHSH violation rules out the stated Bell-local model. Within quantum theory, local measurements on a separable state give $E(\mathbf{a}, \mathbf{b}) = \sum_k p_k A_k(\mathbf{a}) B_k(\mathbf{b})$, where $A_k = \mathrm{Tr}(\rho_A^{(k)} A)$ and $B_k = \mathrm{Tr}(\rho_B^{(k)} B)$ lie in $[-1, 1]$. They obey the same CHSH bound, so a violation certifies entanglement across the declared cut. The numerical CHSH value is not a general entanglement measure, and nonviolation does not prove separability.

Example 2.4 (A mixed-state separation between entanglement and CHSH)**.** Consider the two-qubit Werner family [6], with parameter $0 \le p \le 1$ defined here as the singlet weight:

$$\rho_W(p) = p|\Psi^-\rangle\langle\Psi^-| + (1 - p)\frac{\mathbb{1}_4}{4}. \tag{2.21}$$

The spin flip leaves ρ_W invariant. Its eigenvalues, and thus the r_i in equation (1.30), are $(1+3p)/4$ and three copies of $(1-p)/4$. Consequently

$$C(\rho_W) = \max\left(0, \frac{3p-1}{2}\right). \tag{2.22}$$

The maximally mixed part contributes zero to traceless Pauli correlations, so $E_W(\mathbf{a},\mathbf{b}) = -p\,\mathbf{a}\cdot\mathbf{b}$. Optimizing Alice's directions first and then the angle between Bob's directions gives

$$\begin{aligned}\max_{\text{Pauli settings}} |B_{\text{CHSH}}| &= p \max_{\mathbf{b},\mathbf{b}'}(\|\mathbf{b}-\mathbf{b}'\| + \|\mathbf{b}+\mathbf{b}'\|) \\ &= 2\sqrt{2}\,p. \end{aligned} \tag{2.23}$$

The two norms have squared sum four, so their sum is at most $2\sqrt{2}$, attained for perpendicular $\mathbf{b},\mathbf{b}'$. Hence $1/3 < p \le 1/\sqrt{2}$ gives entangled states with no CHSH violation under any single-copy Pauli settings. For example, $p = 1/2$ gives $C = 1/4$ but the maximum CHSH magnitude is $\sqrt{2}$. This statement does not address other Bell inequalities, local filtering, or collective measurements on several copies.

2.4 Operational ingredients of a Bell test

Electronic Bell tests are motivated both by solid-state quantum-information processing and by quantum electronic optics. In either setting the abstract spin experiment must be translated into a device that prepares a coherent two-level pair, implements independent local basis rotations, and converts the two outcomes into experimentally resolvable currents or counts.

A mesoscopic Bell test requires three ingredients:

1. an entangler that prepares a controlled pair state, usually through an interaction or an effective many-body process;
2. correlated binary measurements, for example Stern–Gerlach spin analysis or a pseudospin-dependent beam splitter followed by counting;
3. tunable analyzer directions $\mathbf{a}, \mathbf{a}', \mathbf{b}, \mathbf{b}'$.

To interpret the measured number as the CHSH correlator, one must also define the trial ensemble and how every accepted trial produces a binary outcome on each side. An input herald or a fixed one-pair preparation can define that ensemble before the analyzer settings are chosen. Discarding detection events according to the later settings or outcomes can invalidate the bound for the remaining sample. A trusted-device entanglement witness may use a justified source and detector model; a Bell nonlocality test must additionally justify locality, setting independence, and its treatment of losses. Current-noise data require a further conversion to these probabilities, developed in the next section.

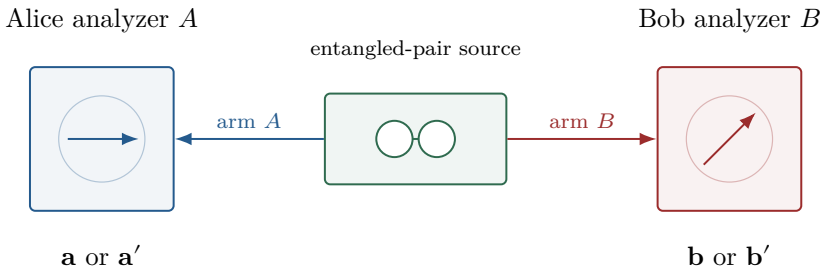


Figure 2.3: A central entangler sends one carrier to each independently controlled local analyzer. The shaded devices are aligned for readability; their dial arrows indicate analyzer settings, not propagation directions. The outward arm arrows show propagation from the source.

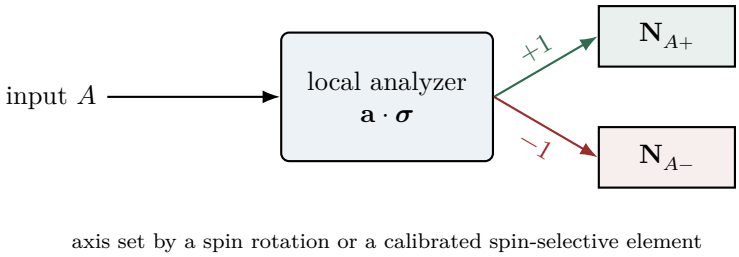


Figure 2.4: Operational two-output spin analyzer: a carrier enters arm A, the measured component is $\mathbf{a}\cdot\boldsymbol{\sigma}$, and the output counts $\mathbf{N}_{A+}, \mathbf{N}_{A-}$ resolve eigenvalues $+1, -1$. The lines label channels, not calculated particle trajectories. Stern–Gerlach splitting uses a magnetic-field gradient acting on a magnetic moment; mesoscopic realizations instead require a calibrated spin-selective transmission and readout.

3 Mesoscopic spin entanglers and current-noise detection

3.1 Operational entanglement resides in accessible modes

The practical target is not arbitrary background entanglement but a deliberately prepared, tunable state between accessible subsystems. One implementation splits a superconducting Cooper pair into two spatial leads by crossed Andreev reflection (CAR). With a fixed fermionic creation-operator order, the emitted spin singlet at energies E and $-E$ is

$$|\Psi_E^-\rangle = \frac{1}{\sqrt{2}}\left[c_{1\uparrow}^\dagger(E)c_{2\downarrow}^\dagger(-E) - c_{1\downarrow}^\dagger(E)c_{2\uparrow}^\dagger(-E)\right]|0\rangle. \tag{3.1}$$

Here leads 1 and 2 are the physical subsystems and $\uparrow, \downarrow$ denote physical spin. We assume singlet s-wave pairing, spin-independent propagation, and suppression or explicit rejection of events in which both electrons enter the same lead. CAR alone does not guarantee these preparation and detection conditions [8]. The state displayed above is the normalized spin state conditional on one electron in each lead, with any common orbital factor suppressed. For this expression, take discrete normalized energy or narrow wave-packet modes; continuum normalization is introduced below.

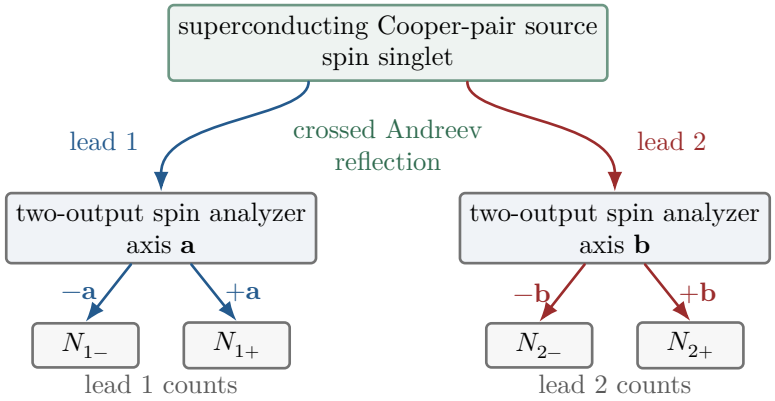


Figure 3.1: Crossed-Andreev entangler with local spin analysis. A superconducting pair is split between leads 1 and 2; ideal analyzers (for example, complementary spin filters) oriented along $\mathbf{a}$ and $\mathbf{b}$ resolve the corresponding $\pm$ outcomes.

A first-quantized antisymmetrized wavefunction can look like a two-term entangled state even when its second-quantized form is a single Slater determinant, for example

$$c_{1\uparrow}^\dagger c_{2\downarrow}^\dagger|0\rangle. \tag{3.2}$$

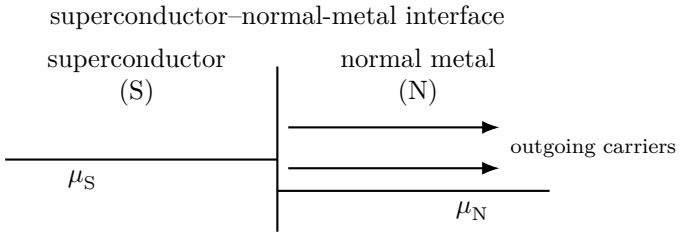


Figure 3.2: Superconductor–normal-metal interface with chemical potentials μ_S and μ_N. The two black arrows indicate carrier flow from the interface into the normal region; their vertical separation is only a visual guide.

The formal particle coordinates in the determinant are not physical subsystem labels. Exchange antisymmetry is kinematics; an operational entanglement claim must instead specify accessible modes, here the spatial leads and their local spin operations. A general correlated fermionic state need not be a single Slater determinant; a sum of determinants is generally necessary.

There is a second operational restriction. If local operations conserve particle number and no shared phase reference is available, coherence between different local number sectors cannot be used as freely as spin coherence within the one-electron sector. For example, the mode state $(|1\rangle_1|0\rangle_2+|0\rangle_1|1\rangle_2)/\sqrt{2}$ is entangled in the mode tensor product, but its locally number-dephased state is a separable mixture under those restrictions. This does not erase all exchange effects or imply that mode entanglement is meaningless. The accessible operations and resources must be stated [7]. The split spin singlet avoids this particular obstacle because each party has exactly one electron.

3.2 From binary counts to a spin correlator

For a spin analyzer along $\mathbf{a}$, define

$$P_\alpha(\mathbf{a}) = \frac{\mathbb{1} + \alpha\, \mathbf{a}\cdot\boldsymbol{\sigma}}{2}, \qquad \alpha = \pm 1. \tag{3.3}$$

If $|\psi\rangle = \alpha_+|+\mathbf{a}\rangle + \alpha_-|-\mathbf{a}\rangle$, then

$$\langle\psi \mid \mathbf{a}\cdot\boldsymbol{\sigma} \mid \psi\rangle = |\alpha_+|^2 - |\alpha_-|^2. \tag{3.4}$$

With measured counts N_+ and N_-, the experimental estimator is

$$\langle\mathbf{a}\cdot\boldsymbol{\sigma}\rangle_{\text{meas}} = \frac{N_+ - N_-}{N_+ + N_-}. \tag{3.5}$$

For two analyzers, expand a normalized state in their outcome bases,

$$|\Psi\rangle = \sum_{\alpha,\beta=\pm1} \alpha_{\alpha\beta}|\alpha\mathbf{a}\rangle_1|\beta\mathbf{b}\rangle_2. \tag{3.6}$$

The binary correlator is

$$\begin{aligned} E(\mathbf{a},\mathbf{b}) &= \langle\Psi \mid (\mathbf{a}\cdot\boldsymbol{\sigma})\otimes(\mathbf{b}\cdot\boldsymbol{\sigma}) \mid \Psi\rangle \\ &= |\alpha_{++}|^2 + |\alpha_{--}|^2 - |\alpha_{+-}|^2 - |\alpha_{-+}|^2. \end{aligned} \tag{3.7}$$

For M_{pair} registered coincidence trials, let $N_{\alpha\beta}$ be the accumulated number of paired outcomes. The estimator is

$$E_{\text{est}}(\mathbf{a},\mathbf{b}) = \frac{N_{++} + N_{--} - N_{+-} - N_{-+}}{\sum_{\alpha,\beta=\pm1} N_{\alpha\beta}}. \tag{3.8}$$

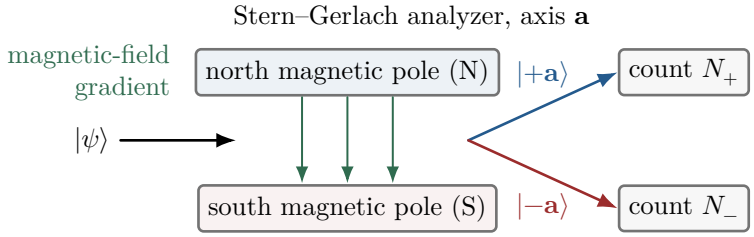


Figure 3.3: Single-particle Stern–Gerlach analyzer. The magnetic-field gradient separates the eigenstates of $\mathbf{a}\cdot\boldsymbol{\sigma}$ into channels counted by N_+ and N_-.

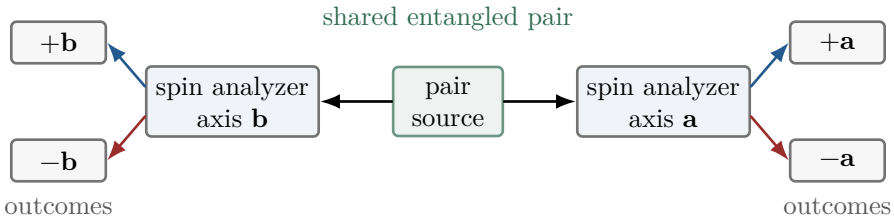


Figure 3.4: Correlated two-arm measurement. The left and right beam splitters analyze along **b** and **a**, respectively, and their four outgoing branches represent the four joint outcomes. The left–right placement is only a drawing convention; subsystem labels follow the arm indices.

outcome	eigenvalue	count
$\lvert+\mathbf{a}\rangle$	$+1$	N_+
$\lvert-\mathbf{a}\rangle$	-1	N_-

Table 3.1: Single-spin outcomes, eigenvalues, and measured counts.

joint outcome	product value	coincidence count
$(+\mathbf{a},+\mathbf{b})$	$+1$	N_{++}
$(+\mathbf{a},-\mathbf{b})$	-1	N_{+-}
$(-\mathbf{a},+\mathbf{b})$	-1	N_{-+}
$(-\mathbf{a},-\mathbf{b})$	$+1$	N_{--}

Table 3.2: Joint outcomes and coincidence counts. The middle column is the product of the two binary eigenvalues.

The denominator is M_{pair}, not M_{pair}^2: it counts paired events. In a large, representative sample the estimator approaches $E = \sum_{\alpha\beta} \alpha\beta p_{\alpha\beta}$, where $p_{\alpha\beta} = \mathrm{Tr}[\rho\, P_\alpha(\mathbf{a}) \otimes P_\beta(\mathbf{b})]$. Coincidence selection must not depend on the settings or outcomes in a way that biases the sample. A noise-based reconstruction under a source model is not, by itself, a loophole-free Bell test [9].

3.3 Finite counting time and connected current noise

Let $T \equiv t_{\text{det}}$ be a detection window satisfying

$$\tau \ll T \ll \Delta T, \tag{3.9}$$

where τ is a single-pair coherence time and ΔT is the typical separation between emitted pairs. Most untriggered windows are empty; the probability of two or more pairs is negligible rather than identically zero. Detection windows on the two arms are aligned after allowing for the flight times. A triggered, exactly-one-pair experiment has different number statistics and must be distinguished from this dilute stationary source.

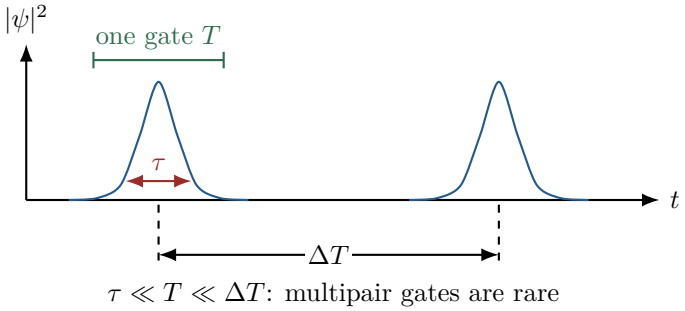


Figure 3.5: Time scales for resolving individual emitted pairs. Each wave packet has coherence width τ, neighboring emissions are separated by ΔT, and the detection gate obeys $\tau \ll T = t_{\text{det}} \ll \Delta T$. The packets are schematic; the gate shown happens to contain a pair.

Use $n_{i\alpha} \in \{0, 1\}$ for a single-window count, distinct from the accumulated $N_{\alpha\beta}$. In an ideal at-most-one-pair model, let q be the probability that a window contains a detected pair and let $p_{\alpha\beta}$ be its conditional outcome distribution. Then

$$\begin{aligned} \langle n_{1\alpha} n_{2\beta} \rangle &= q p_{\alpha\beta}, \qquad \langle n_{1\alpha} \rangle = q p_{1\alpha}, \qquad \langle n_{2\beta} \rangle = q p_{2\beta}, \\ K_{\alpha\beta} &\equiv \langle \delta n_{1\alpha} \delta n_{2\beta} \rangle = q p_{\alpha\beta} - q^2 p_{1\alpha} p_{2\beta}. \end{aligned} \tag{3.10}$$

Here $p_{1\alpha} = \sum_\beta p_{\alpha\beta}$ and similarly for arm 2. The exact relation, also useful beyond this simple source model, is

$$E = \frac{\sum_{\alpha\beta} \alpha\beta [K_{\alpha\beta} + \langle n_{1\alpha} \rangle \langle n_{2\beta} \rangle]}{\sum_{\alpha\beta} [K_{\alpha\beta} + \langle n_{1\alpha} \rangle \langle n_{2\beta} \rangle]}. \tag{3.11}$$

It represents a pair correlator when the count products represent the exclusive pair outcomes. The numerator reduces to a signed covariance if one local mean spin vanishes. The denominator still includes the means: $\sum K_{\alpha\beta} = q(1 - q)$, whereas the full coincidence sum is q. For $q = 1$ the covariance sum is zero. For $q \ll 1$, discarding the q^2 terms is a controlled approximation.

An alternative is an ideal stream of independent Poisson pair emissions at rate R. Splitting that stream among outcomes with probabilities $p_{\alpha\beta}$ gives $K_{\alpha\beta} = RTp_{\alpha\beta}$, after the same-pair pulse is fully integrated. Here the means' products describe accidental pairs from distinct emissions, and connected noise isolates the same-pair rate. These two statistical models explain both the usefulness and the limits of replacing coincidences by noise.

The count operator associated with the ideal electron detector is $\mathbf{N}_{i\alpha} = e^{-1}\int_0^T \mathbf{I}_{i\alpha}(t)\,\mathrm{d}t$, where $\mathbf{I}$ is $e > 0$ times the outgoing electron flux. Its measured outcomes are the $n_{i\alpha}$. Define signed currents

$$\mathbf{I}_{1,\mathbf{a}} = \mathbf{I}_{1+} - \mathbf{I}_{1-}, \qquad \mathbf{I}_{2,\mathbf{b}} = \mathbf{I}_{2+} - \mathbf{I}_{2-}. \tag{3.12}$$

For a stationary source, use the *two-sided ordered* connected spectrum throughout this chapter:

$$C_{\alpha\beta}(u) = \langle \delta\mathbf{I}_{1\alpha}(u)\delta\mathbf{I}_{2\beta}(0)\rangle, \qquad S_{\alpha\beta}(\omega) = \int_{-\infty}^{\infty} \mathrm{d}u\, \mathrm{e}^{\mathrm{i}\omega u} C_{\alpha\beta}(u). \tag{3.13}$$

ω is angular frequency. C has units A^2 and S has units $\mathrm{A}^2\,\mathrm{s}$; the inverse transform includes $1/(2\pi)$. A symmetrized spectrum defined with one half of the anticommutator is a different object at finite frequency. A frequently used convention omits that half; for a real commuting zero-frequency cross correlator it reports $\widetilde{S}(0) = 2S(0)$. All prefactors below use S, not $\widetilde{S}$. For ideal counts on separate outgoing arms, the integrated observables commute. Actual detector ordering, bandwidth and any equilibrium background must be calibrated before identifying its output with this spectrum.

The finite detector gate enters through the elementary Fourier integrals

$$\int_0^T \mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t = \frac{\mathrm{e}^{-\mathrm{i}\omega T} - 1}{-\mathrm{i}\omega}, \qquad \int_0^T \mathrm{e}^{\mathrm{i}\omega t'}\,\mathrm{d}t' = \frac{\mathrm{e}^{\mathrm{i}\omega T} - 1}{\mathrm{i}\omega}, \tag{3.14}$$

$$\left|\int_0^T \mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t\right|^2 = \frac{2[1-\cos(\omega T)]}{\omega^2} = \frac{4\sin^2(\omega T/2)}{\omega^2}. \tag{3.15}$$

Thus a finite gate applies a low-frequency window; it does not amount to setting $\omega = 0$ before the integration.

The double time integral becomes

$$\langle \delta\mathbf{N}_{1\alpha}\delta\mathbf{N}_{2\beta}\rangle = \frac{1}{2\pi e^2}\int_{-\infty}^{\infty} \mathrm{d}\omega\, \frac{2[1-\cos(\omega T)]}{\omega^2} S_{\alpha\beta}(\omega) \tag{3.16}$$

$$= \frac{1}{e^2}\int_{-\infty}^{\infty} \mathrm{d}\omega\, \frac{2}{\pi\omega^2}\sin^2\!\left(\frac{\omega T}{2}\right) S_{\alpha\beta}(\omega). \tag{3.17}$$

In the long-window limit $T \gg \tau \simeq \hbar/(eV)$,

$$\frac{2}{\pi\omega^2}\sin^2\!\left(\frac{\omega T}{2}\right) \longrightarrow T\,\delta(\omega), \qquad \langle \delta\mathbf{N}_{1\alpha}\delta\mathbf{N}_{2\beta}\rangle \longrightarrow \frac{T}{e^2} S_{\alpha\beta}(0). \tag{3.18}$$

The arrow is a distributional low-frequency approximation: $S(\omega)$ must vary little over the main detector width $1/T$. It does not make a finite-window count into a frequency-resolved measurement. In the dilute-pair limit (or the independent Poisson model), and after removing calibrated detector/equilibrium backgrounds, one obtains

$$E(\mathbf{a},\mathbf{b}) \simeq \frac{S_{++}(0) + S_{--}(0) - S_{+-}(0) - S_{-+}(0)}{\sum_{\alpha\beta} S_{\alpha\beta}(0)}. \tag{3.19}$$

For general count statistics the mean-current products in equation (3.11) must instead be retained; in its long-window form these contribute $T\langle I_{1\alpha}\rangle\langle I_{2\beta}\rangle$ beside $S_{\alpha\beta}(0)$. Raw cross noise can be negative and is not a probability. Source purity, local-pair leakage, dark counts, losses and setting-dependent efficiencies can all spoil this reconstruction. Subtracting a background is part of a stated detector/source model, not a substitute for the assumptions of a Bell inequality.

Exercise 3.1. An ideal source delivers exactly one singlet pair per gate. Why does the ratio of signed to unsigned count covariances fail? *Answer:* $\sum_{\alpha\beta} K_{\alpha\beta} = \mathrm{Cov}(1,1) = 0$. The full coincidence denominator is 1, and the signed numerator equals E because the two local mean spins vanish. Adding the disconnected terms restores the correct count formula.

3.4 Weak-tunnelling model for the emitted singlet

We now specify an effective scattering model, so that the noise prefactors can be checked independently of a microscopic contact calculation. Measure electron energies from $\mu_S = 0$, take the normal reservoirs at $\mu_N = -eV$ with $V > 0$, and assume zero temperature and $eV \ll \Delta_{SC}$. Emitted partners have energies $E, -E$. Both must lie above the normal Fermi sea, so

$$E > -eV, \quad -E > -eV \quad \Longrightarrow \quad -eV < E < eV. \tag{3.20}$$

The window is $2eV$ wide, although the reservoir potential difference is eV. We assume a flat, spin-independent pair amplitude in this interval, negligible same-lead emission and no energy-dependent spin rotation. These are model assumptions; a real contact has a device-dependent amplitude [8].

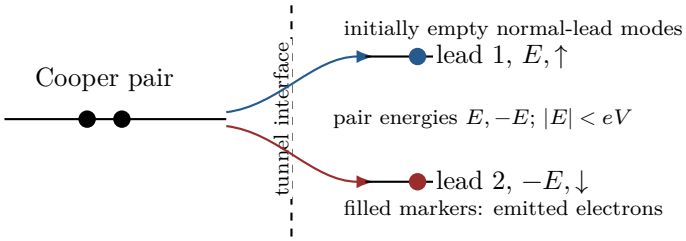


Figure 3.6: Pair-tunnelling process in the bias window. The two particles populate outgoing modes with opposite spins, while Pauli blocking restricts the available states in the normal region to $-eV < E < eV$. Only the $\uparrow_1\downarrow_2$ alternative is drawn; the singlet includes its coherent partner with the minus sign in equation (3.1).

For a finite observation time $\mathcal{T}$, discretize the outgoing scattering energies with spacing $\delta E = h/\mathcal{T}$. With canonical discrete operators $c_{i\sigma,k}$ and $\bar{k}$ denoting energy $-E_k$, a normalized weak-pairing state is

$$|\Psi_{\mathcal{T}}\rangle = \prod_{k:\,|E_k|<eV} \frac{(1 + \Gamma c^\dagger_{1\uparrow,k} c^\dagger_{2\downarrow,\bar{k}})(1 - \Gamma c^\dagger_{1\downarrow,k} c^\dagger_{2\uparrow,\bar{k}})}{1 + |\Gamma|^2} |0\rangle. \tag{3.21}$$

Here $|0\rangle$ is the reference sea, empty in the selected emission modes, and Γ is a dimensionless complex scattering amplitude with $|\Gamma| \ll 1$. Each factor acts on distinct modes. Its single-pair sector is the required singlet, while normalization and rare double-pair sectors are explicit. The full steady source is not a normalized state containing only one pair. The probability for one pair in an energy bin is $2|\Gamma|^2 + O(|\Gamma|^4)$.

The microscopic relation between Γ and a tunnelling Hamiltonian is not fixed here. It requires a scattering or time-dependent transition calculation, including densities of states and contact geometry. In particular, bias energy alone does not justify inserting the nondegenerate stationary-perturbation denominator $-2eV$ and identifying $\Gamma = \gamma/(2eV)$.

In continuum normalization, $\{c_{i\alpha}(E), c^\dagger_{j\beta}(E')\} = \delta_{ij}\delta_{\alpha\beta}\delta(E - E')$; thus $c(E)$ has units of energy$^{-1/2}$. To leading order in this model,

$$\langle c^\dagger_{i\alpha}(E') c_{i\beta}(E)\rangle = |\Gamma|^2[1 + O(|\Gamma|^2)]\delta_{\alpha\beta}\delta(E - E'), \tag{3.22}$$

$$\langle c_{2\beta}(E') c_{1\alpha}(E)\rangle = \Gamma[1 + O(|\Gamma|^2)] J_{\alpha\beta}\delta(E + E'), \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{3.23}$$

All displayed energies lie in $W = (-eV, eV)$. In this ideal model the detector collects that outgoing emission band, with no filled-sea contribution:

$$\mathbf{I}_{i\alpha}(t) = \frac{e}{h} \int_W \mathrm{d}E \int_W \mathrm{d}E'\, \mathrm{e}^{\mathrm{i}(E'-E)t/\hbar} c^\dagger_{i\alpha}(E') c_{i\alpha}(E). \tag{3.24}$$

It is e times electron flux toward the detector. The conventional electric current of electrons in that orientation is $-\mathbf{I}$. A full lead-current measurement also contains equilibrium and incoming contributions; identifying its excess noise with this filtered model requires the background and bandwidth conditions stated above.

The energy delta function removes the time phase and leaves the window width:

$$\langle \mathbf{I}_{i\alpha} \rangle = \frac{e}{h} \int_{-eV}^{eV} \mathrm{d}E\, |\Gamma|^2 = \frac{2|\Gamma|^2 e^2 V}{h} [1 + O(|\Gamma|^2)]. \tag{3.25}$$

Each arm contains two spin outcomes. The pair rate and mean emission separation, to this order, are therefore

$$R = \frac{\langle \mathbf{I}_{i+} + \mathbf{I}_{i-} \rangle}{e} = \frac{4|\Gamma|^2 eV}{h}, \qquad \Delta T = R^{-1}. \tag{3.26}$$

The units are transparent: e^2V/h is a current and eV/h is a rate. For $\tau = \hbar/(eV)$, $R\tau = 2|\Gamma|^2/\pi \ll 1$, so the two conditions $\tau \ll T$ and $RT \ll 1$ can hold together. The vanishing current as $V \to 0$ is consistent with the shrinking emission window, although the separation of detector time scales must then be reconsidered.

3.5 Spin-resolved cross noise

For analyzers restricted to the x–z plane, use the real spinor rotation

$$\begin{pmatrix} c_{i+}(\theta_i) \\ c_{i-}(\theta_i) \end{pmatrix} = \begin{pmatrix} \cos(\theta_i/2) & \sin(\theta_i/2) \\ -\sin(\theta_i/2) & \cos(\theta_i/2) \end{pmatrix} \begin{pmatrix} c_{i\uparrow} \\ c_{i\downarrow} \end{pmatrix}. \tag{3.27}$$

Rotating the state and rotating the measured current operator are equivalent provided the same basis convention is used consistently. This matrix measures $\mathbf{a}_i = (\sin\theta_i, 0, \cos\theta_i)$, since $R_i^\dagger \sigma_z R_i = \sin\theta_i \sigma_x + \cos\theta_i \sigma_z$.

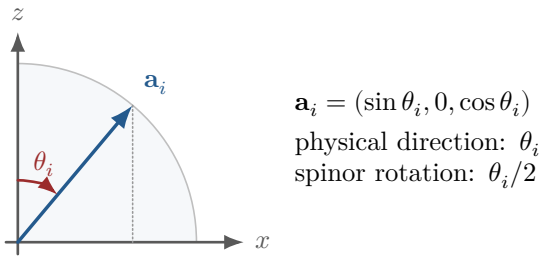


Figure 3.7: Spin-analyzer coordinates. Positive θ_i rotates the $+z$ axis toward $+x$, consistently with equation (3.27).

For the rotated pair amplitude, the matrix is $R_1 J R_2^{\mathsf{T}}$; its $++$ element is $-\sin[(\theta_1 - \theta_2)/2]$. Define the filtered field $\psi_{i\alpha}(t) = h^{-1/2} \int_W \mathrm{d}E\, \mathrm{e}^{-\mathrm{i}Et/\hbar} c_{i\alpha}(E)$, so that $\mathbf{I} = e\psi^\dagger\psi$. The connected Wick contraction for distinct arms gives

$$C_{++}(u) = e^2 |\langle \psi_{2+}(0)\psi_{1+}(u)\rangle|^2 - e^2 |\langle \psi_{1+}^\dagger(u)\psi_{2+}(0)\rangle|^2. \tag{3.28}$$

The normal cross-arm coherence in the second term vanishes in this model. The positive first term comes from the anomalous pair amplitude, with the fermionic reordering already included. Its energy integral is

$$K(u) = \int_{-eV}^{eV} \mathrm{d}E\, \mathrm{e}^{-\mathrm{i}Eu/\hbar} = \frac{2\hbar \sin(eVu/\hbar)}{u}, \qquad K(0) = 2eV. \tag{3.29}$$

Consequently,

$$C_{++}(t-t') = \frac{|\Gamma|^2 e^2}{h^2} \sin^2\left(\frac{\theta_1-\theta_2}{2}\right) \frac{4\hbar^2}{(t-t')^2} \sin^2\left(\frac{eV(t-t')}{\hbar}\right) [1+O(|\Gamma|^2)]. \tag{3.30}$$

This is a connected correlation from the outset. Its finite value at $t = t'$ is set by $K(0)$, and its characteristic width is $\tau = \hbar/(eV)$. The first zeros occur at $|t-t'| = \pi\tau$. A non-flat energy filter changes the pulse shape to the squared Fourier transform of its pair amplitude.

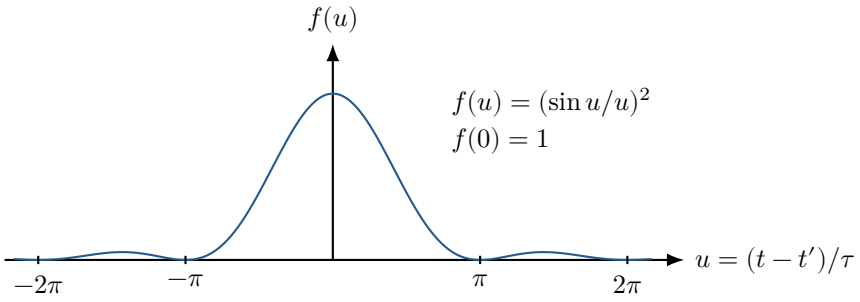


Figure 3.8: Exact normalized pulse for the flat bias-window model: $f(u) = |K(\tau u)/K(0)|^2$. There are no fitted parameters; the zeros at $u = \pm\pi, \pm 2\pi$ check the time-axis convention. The physical amplitude and analyzer factor appear in equation (3.30).

Using $\int_{-\infty}^{\infty} \sin^2(ax)\, dx/x^2 = \pi|a|$ gives $\int |K(u)|^2\, du = 4\pi\hbar eV = 2heV$. Thus, with exactly the spectral convention of equation (3.13),

$$S_{++}(0) = \int_{-\infty}^{\infty} du\, C_{++}(u) = S_0 \sin^2\left(\frac{\theta_1-\theta_2}{2}\right) [1+O(|\Gamma|^2)], \tag{3.31}$$

$$S_0 = \frac{2|\Gamma|^2 e^3 V}{h} = \frac{e^2 R}{2}. \tag{3.32}$$

There is no extra factor of 2 in this definition. A convention that reports $\widetilde{S} = 2S$ would instead give $\widetilde{S}_0 = 4|\Gamma|^2 e^3 V/h$. No disconnected constant is integrated over infinite time: connectedness was enforced before taking this limit.

Changing one analyzer outcome shifts its angle by π, so

$$S_{++} = S_{--} = S_0 \sin^2\left(\frac{\theta_1-\theta_2}{2}\right), \tag{3.33}$$

$$S_{+-} = S_{-+} = S_0 \cos^2\left(\frac{\theta_1-\theta_2}{2}\right), \tag{3.34}$$

$$\sum_{\alpha,\beta} S_{\alpha\beta} = 2S_0. \tag{3.35}$$

Substitution into equation (3.19) yields

$$E(\mathbf{a},\mathbf{b}) = -\cos(\theta_1-\theta_2) = -\mathbf{a}\cdot\mathbf{b} \tag{3.36}$$

for coplanar analyzers, to the retained order in $|\Gamma|^2$. In fact $S_{\alpha\beta}(0) = e^2 R p_{\alpha\beta}$ with $p_{\alpha\beta} = [1 - \alpha\beta\mathbf{a}\cdot\mathbf{b}]/4$. This explicitly connects the scattering result to independent pair counting: the sum is e^2R, not an arbitrary noise normalization. Four resolved spin-noise components, measured simultaneously or in separate calibrated configurations, reconstruct the singlet correlator under the stated source and detection conditions.

Exercise 3.2. Check equal and opposite analyzer directions. *Answer:* For $\mathbf{a} = \mathbf{b}$, $S_{++} = S_{--} = 0$ and $E = -1$; for $\mathbf{a} = -\mathbf{b}$, $S_{+-} = S_{-+} = 0$ and $E = +1$. The sum stays e^2R in both cases. Positive pair cross noise by itself is not an entanglement witness; the setting dependence supplies the spin test.

4 Orbital and Nambu pseudospin entanglement

The same two-dimensional Hilbert-space algebra describes several physically different qubits. The measurement device is determined by the encoding, not by the appearance of a Pauli matrix:

encoding	ordered basis	Pauli vector	direct readout
physical spin	$(\uparrow,\downarrow)$	$\boldsymbol{\sigma}$	spin-resolved counts
orbital channel	$(\uparrow,\downarrow)_{\mathrm{orb}}$	$\boldsymbol{\mu}$	channel counts
Nambu charge	(e,h)	$\boldsymbol{\tau}$	signed charge

The arm order in the bipartite Hilbert space is always $1 \otimes 2$. An electron–hole pair can carry an *orbital* qubit on each carrier, or the electron/hole alternatives can themselves be the qubit. These are different partitions of the available degrees of freedom.

4.1 Orbital two-level systems

The two-level algebra of spin can be carried by spatial channels instead. For two regions $1, 2$ with ordered orbital modes $\uparrow,\downarrow$, an orbital-entangled pair amplitude takes the form

$$|\Psi_{\mathrm{orb}}\rangle = \frac{1}{\sqrt{2}}\left(c^\dagger_{1\uparrow}c^\dagger_{2\uparrow} + c^\dagger_{1\downarrow}c^\dagger_{2\downarrow}\right)|0\rangle. \tag{4.1}$$

The arrows in this section are orbital or lead labels, not physical spin. The corresponding Pauli vector is therefore $\boldsymbol{\mu}$, not $\boldsymbol{\sigma}$. The modes in this expression are orthonormal wave-packet modes, and the state is conditioned on one excitation per region. Any unshown physical-spin or energy state must be common to the two orbital alternatives, or its distinguishing information must be retained in the description. Equal amplitudes and a fixed relative phase give the displayed Bell state. The two superconducting contacts in the schematic must therefore be phase coherent; two unrelated sources with a fluctuating relative phase do not prepare this pure orbital state.

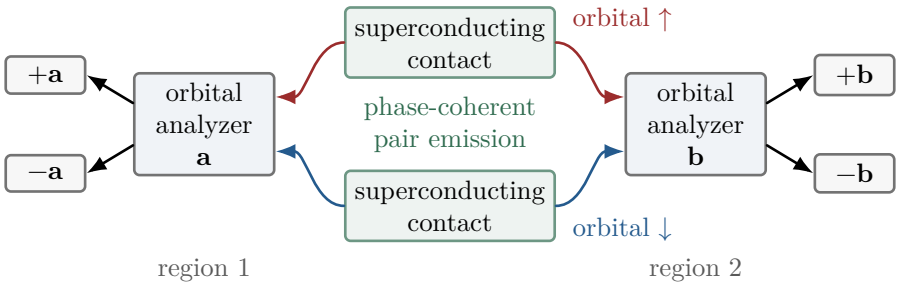


Figure 4.1: Orbital-pair entangler and local analyzers. Phase-coherent superconducting contacts supply the alternatives "both particles in the upper orbital" and "both particles in the lower orbital"; local beam splitters analyze the orbital pseudospins along **a** and **b**. Equal amplitudes and zero relative phase are assumed in the displayed state.

When the relevant orbital rotations are real and coplanar, the analyzer basis can be written

$$\begin{pmatrix} c_{i+}(\theta_i) \\ c_{i-}(\theta_i) \end{pmatrix} = \begin{pmatrix} \cos(\theta_i/2) & \sin(\theta_i/2) \\ -\sin(\theta_i/2) & \cos(\theta_i/2) \end{pmatrix} \begin{pmatrix} c_{i\uparrow} \\ c_{i\downarrow} \end{pmatrix}. \tag{4.2}$$

In a general interferometer, propagation phases make an arbitrary SU(2) orbital rotation necessary; the real matrix in equation (4.2) is not universal.

4.1.1 Electron–hole orbital entanglement from a filled sea

Consider two regions in the integer quantum Hall regime coupled by a quantum point contact (QPC). In this setting an edge channel is chiral: at a given boundary it propagates in a fixed direction set by the magnetic field and edge orientation. Two co-propagating channels provide the two orbital labels. They are not the counterpropagating time-reversed spin partners of a quantum spin Hall edge. The microscopic spin character of the channels and whether a proposed mixer couples them must be checked in a real device. The two-channel QPC is a theoretical electron–hole orbital entangler [10]. Distinguish the empty-electron vacuum $|\mathrm{vac}\rangle$ from the biased filled reference sea

$$|0_V\rangle = \left[\prod_{\alpha}\prod_{E<eV} c^\dagger_{1\alpha}(E)\right]\left[\prod_{\alpha}\prod_{E<0} c^\dagger_{2\alpha}(E)\right]|\mathrm{vac}\rangle, \qquad \alpha\in\{\uparrow,\downarrow\}. \tag{4.3}$$

The products are symbolic finite-volume products; region 1 is filled up to eV, whereas region 2 is filled only up to the zero-energy Fermi level. The QPC tunnelling operator is schematically

$$H_{\mathrm{QPC}} = \sum_{\alpha=\uparrow,\downarrow}\left(\eta_\alpha c^\dagger_{2\alpha}c_{1\alpha} + \eta^*_\alpha c^\dagger_{1\alpha}c_{2\alpha}\right). \tag{4.4}$$

Here η_α is an energy-valued matrix element between discrete modes; the displayed expression is a modewise schematic. The corresponding dimensionless transfer amplitudes $t_\alpha(E)$ require the scattering normalization and need not equal η_α. Taking them equal and flat, $t_\uparrow = t_\downarrow = t_0$, simplifies the illustration below. Within the transport window $0 < E < eV$, the bias selects the transfer from the filled region-1 mode to the empty region-2 mode. To first order, the resulting excitation is

$$|\delta\Psi_{\mathrm{QPC}}\rangle \propto t_0 \sum_{0<E<eV}\left(c^\dagger_{2\uparrow}(E)c_{1\uparrow}(E) + c^\dagger_{2\downarrow}(E)c_{1\downarrow}(E)\right)|0_V\rangle. \tag{4.5}$$

Performing the particle–hole transformation in region 1 relative to this filled sea,

$$d^\dagger_{1\alpha}(E) \equiv c_{1\alpha}(E), \qquad \alpha\in\{\uparrow,\downarrow\}, \tag{4.6}$$

turns the excitation above the Fermi sea into

$$|\Psi_{eh,\mathrm{orb}}\rangle \propto \sum_{0<E<eV}\left(c^\dagger_{2\uparrow}(E)d^\dagger_{1\uparrow}(E) + c^\dagger_{2\downarrow}(E)d^\dagger_{1\downarrow}(E)\right)|0_V\rangle. \tag{4.7}$$

Here a hole is a vacancy relative to $|0_V\rangle$, not an additional electron in the empty vacuum. In $d^\dagger(E)$, E labels the *removed electron's* energy. Its positive excitation energy measured from the region-1 sea is $eV - E$; the electron in region 2 has excitation energy E. The diagram must therefore place the hole in region 1 and the added electron in region 2. Reordering $c^\dagger_2 d^\dagger_1$ into the arm order $d^\dagger_1 c^\dagger_2$ introduces one common minus sign and changes no relative phase.

Conditional on one transferred pair with a common orbital-independent wave-packet profile, the normalized orbital state is

$$|\psi_{eh,\mathrm{orb}}\rangle = \frac{t_\uparrow|\uparrow\rangle_{1h}|\uparrow\rangle_{2e} + t_\downarrow|\downarrow\rangle_{1h}|\downarrow\rangle_{2e}}{\sqrt{|t_\uparrow|^2+|t_\downarrow|^2}},$$
$$C = \frac{2|t_\uparrow t_\downarrow|}{|t_\uparrow|^2+|t_\downarrow|^2}. \tag{4.8}$$

The concurrence follows directly from the two Schmidt probabilities. It is unity only for equal nonzero magnitudes. The labels e, h describe the carriers, while the qubits are the two orbital channels. If energy and orbital alternatives are correlated, tracing energy generally produces a mixed orbital state and this pure-state formula no longer applies.

A hole transforms in the conjugate representation of an electron basis change: if $c' = Uc$, then $d'^{\dagger} = Ud^{\dagger}$ and $d' = U^{*}d$. The analyzer convention must account for this conjugation when it includes complex phases. For the real rotation in equation (4.2) the distinction is hidden. No direct electron–electron interaction is required for this weak-QPC preparation; mode partition, filled-sea bookkeeping and conditioning identify the useful two-excitation sector [10].

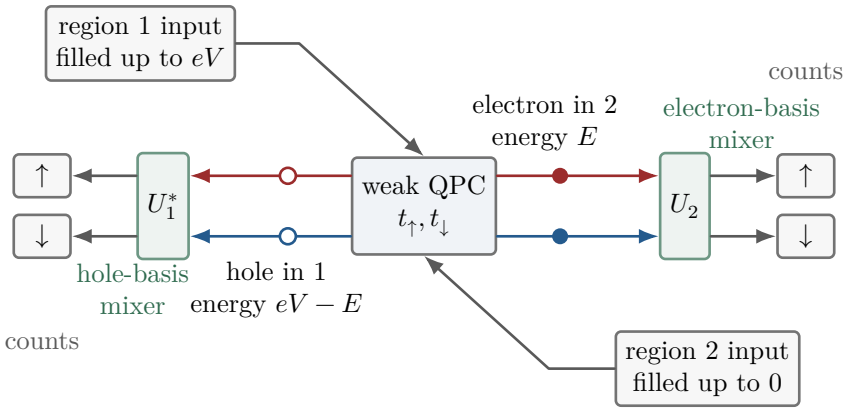


Figure 4.2: Mode-connectivity schematic of the orbital QPC entangler, not a scale drawing of a Hall bar. At $0 < E < eV$ a coherent transfer leaves a hole in arm 1 and adds an electron in arm 2, in either orbital channel. The two colors show coherent alternatives, not two pairs emitted together. Open markers denote holes and filled markers denote added electrons. Green local mixers act before orbital-resolved counting at the outward arrowheads.

Three limitations are important: entanglement must be assigned to physical modes; correlations between the entangled degrees of freedom may be inherited from the preparation process or the reference sea; and complete orbital analyzer rotations are difficult because uncontrolled phases cannot in general be removed by assuming a single x–z plane.

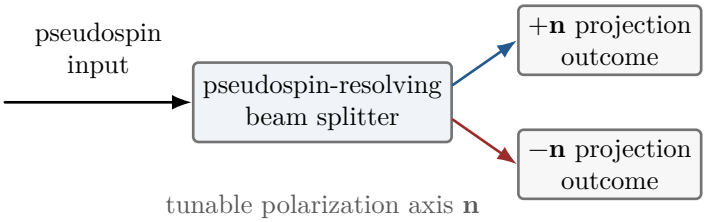


Figure 4.3: Ideal pseudospin-dependent beam splitter. It converts a two-level projection into two spatially resolved outcomes. Incomplete polarization and backscattering make such an analyzer experimentally demanding.

4.2 Nambu electron–hole pseudospin

In Nambu encoding the logical alternatives on each arm are an electron excitation and a hole excitation of an effective channel. Relative to a common zero-temperature sea at $\mu = 0$, use positive excitation energies $E > 0$ and define

$$\gamma_{ie}^{\dagger}(E) = c_i^{\dagger}(E), \qquad \gamma_{ih}^{\dagger}(E) = c_i(-E). \tag{4.9}$$

Using only $E > 0$ avoids double counting the particle–hole redundancy of a Bogoliubov–de Gennes (BdG) description. The effective two-component block and any physical-spin partners are fixed by the device; the Nambu labels are not additional independent spin states.

For matched wave packets and one excitation in each arm, an ideal electron–hole Bell pair is written

$$|\Psi^{-}_{eh}\rangle = \frac{|e\rangle_1|h\rangle_2 - |h\rangle_1|e\rangle_2}{\sqrt{2}} = \frac{\gamma^\dagger_{1e}\gamma^\dagger_{2h} - \gamma^\dagger_{1h}\gamma^\dagger_{2e}}{\sqrt{2}}|0\rangle. \tag{4.10}$$

The operators in the second expression denote normalized wave-packet modes. The minus sign assumes the displayed arm order; a controllable relative phase can be included and compensated by a local phase rotation. An ac-driven QPC provides a theoretical route to this nonlocal component [11]. Its complete output also contains vacuum, local electron–hole pairs and, at stronger drive, multipair sectors. Obtaining the displayed pure Nambu state requires matched amplitudes and no unobserved energy, time or spin information that distinguishes the two alternatives. These conditions must be checked when a noise experiment is interpreted as a two-qubit measurement.

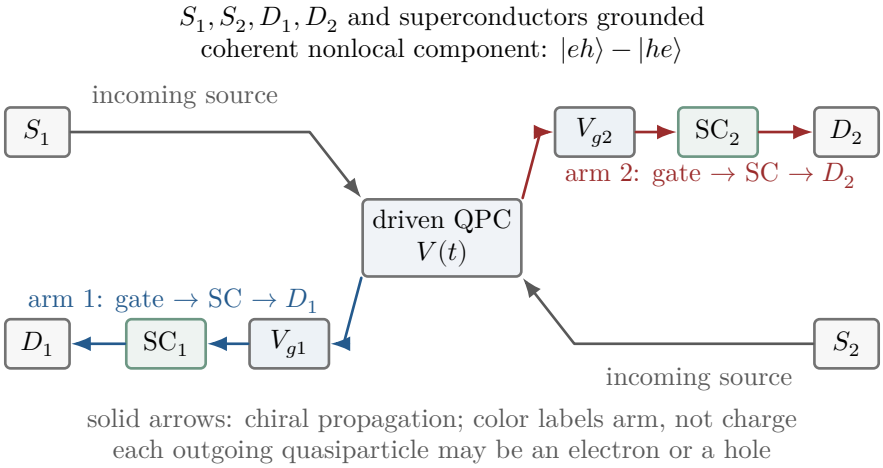


Figure 4.4: Four-port schematic of a driven Nambu-entangler proposal. Incoming sources S_1, S_2 feed the ac-driven QPC. The two outgoing arms encounter a voltage gate and then a grounded superconducting segment, before charge detection at D_1, D_2. The ordered segments implement $U_s U_g$. The drawing specifies the ideal channel connections and does not assert that the full driven output is only the nonlocal Bell pair.

Use the ordered Nambu spinor

$$\boldsymbol{\gamma}_i(E) = \begin{pmatrix} \gamma_{ie}(E) \\ \gamma_{ih}(E) \end{pmatrix}, \qquad \tau_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad q_{\mathrm{N}} = -e\tau_z, \qquad e > 0. \tag{4.11}$$

Local superpositions $\alpha|e\rangle + \beta|h\rangle$ mix charge sectors differing by $2e$. A superconducting phase reference supplies the charge exchange needed to prepare and measure their coherence. Without such a reference, strictly charge-conserving local operations cannot implement arbitrary Nambu rotations; diagonal charge measurements alone do not reveal the off-diagonal coherence. Physical charge conservation applies to the quasiparticles plus condensate, not to the reduced Nambu qubit alone.

Define the ideal detector-oriented quasiparticle signal by

$$\mathbf{I}^{\mathrm{det}}_i(t) = \frac{e}{h}\int_0^\infty \mathrm{d}E \int_0^\infty \mathrm{d}E'\, \mathrm{e}^{\mathrm{i}(E-E')t/\hbar}\boldsymbol{\gamma}^\dagger_i(E)\tau_z\boldsymbol{\gamma}_i(E'). \tag{4.12}$$

The physical electric charge-current operator in the same propagation orientation is $\mathbf{I}_{i,Q} = -\mathbf{I}^{\mathrm{det}}_i$, consistently with $q_{\mathrm{N}} = -e\tau_z$. This positive-excitation expression describes the detected outgoing quasiparticle contribution. A complete lead-current operator includes incoming and filled-sea

terms; interpreting the measured excess noise requires the detector and background calibration of section 3.

For a single Nambu state $|s\rangle = \alpha|e\rangle + \beta|h\rangle$,

$$\langle s \mid \tau_z \mid s\rangle = |\alpha|^2 - |\beta|^2. \tag{4.13}$$

Thus charge readout directly measures the Nambu-z pseudospin. Unlike a spin analyzer that separates two spin terminals, electron and hole contributions already enter one charge-current signal with opposite signs. Consequently, a separate pseudospin beam splitter is unnecessary for the z readout.

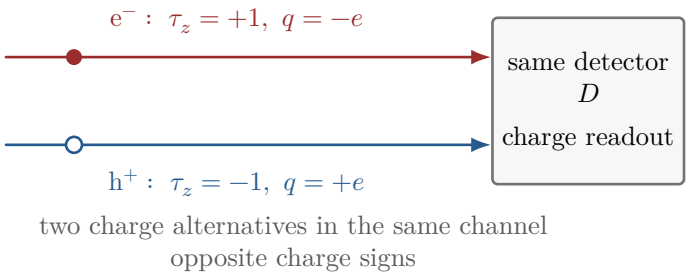


Figure 4.5: Nambu charge readout. A filled marker denotes an electron and an open marker a hole. Both propagate to the same detector terminal, while their opposite charges make one current sensitive to Nambu τ_z.

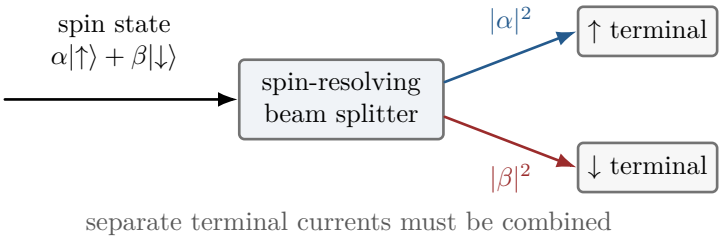


Figure 4.6: Physical-spin readout. A spin-dependent beam splitter converts the two projections into separate output terminals, so forming a signed spin signal requires combining two detector currents.

4.2.1 General Nambu analyzer direction

For

$$\mathbf{a} = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta), \tag{4.14}$$

let $U(\mathbf{a})$ rotate the desired Bloch direction to the measured z axis. The generalized detector signal is

$$\mathbf{I}_i^{\mathrm{det}}(\mathbf{a}, t) = \frac{e}{h}\int_0^\infty \mathrm{d}E \int_0^\infty \mathrm{d}E'\, \mathrm{e}^{\mathrm{i}(E-E')t/\hbar}\boldsymbol{\gamma}_i^\dagger(E)U^\dagger(\mathbf{a})\tau_z U(\mathbf{a})\boldsymbol{\gamma}_i(E'). \tag{4.15}$$

A convenient implementation decomposes the rotation as

$$U(\mathbf{a}) = U_s(-\theta)U_g(-\varphi - \pi/2), \tag{4.16}$$

$$U_s(\theta') = \exp(-\mathrm{i}\theta'\tau_x/2), \qquad \theta' = \frac{2\Delta L_s}{\hbar v}, \tag{4.17}$$

$$U_g(\varphi') = \exp(-\mathrm{i}\varphi'\tau_z/2), \qquad \varphi' = -\frac{2eV_g L_g}{\hbar v}. \tag{4.18}$$

Here L_s is the length of a segment with pairing scale Δ, L_g is a gated length, V_g is its gate voltage, and v is the chiral-mode velocity. These formulas assume a uniform ballistic effective BdG channel with Hamiltonians $H_s = \Delta\tau_x$ and $H_g = -eV_g\tau_z$, transit times $L_s/v, L_g/v$, and a fixed superconducting phase chosen as zero. Δ is an effective induced pairing matrix element with units of energy, not a promise that every chiral channel admits this simple two-level model. For a superconducting phase χ, the conversion axis lies in the x–y plane and must be included in the calibration. The experiment must be able to realize the required signed rotations, or equivalent angles modulo 2π.

Matrix multiplication acts from right to left: the particle encounters the gate U_g first, then the superconducting segment U_s. Direct Pauli algebra gives

$$\begin{aligned} U_s(-\theta)^\dagger \tau_z U_s(-\theta) &= \cos\theta\,\tau_z - \sin\theta\,\tau_y, \\ U(\mathbf{a})^\dagger \tau_z U(\mathbf{a}) &= \sin\theta\cos\varphi\,\tau_x + \sin\theta\sin\varphi\,\tau_y + \cos\theta\,\tau_z = \mathbf{a}\cdot\boldsymbol{\tau}. \end{aligned} \tag{4.19}$$

Thus an arbitrary desired input projection is converted into the measured charge projection. Reversing the physical order would give U_gU_s; the final z rotation would commute with charge readout and could not independently tune the azimuth in this two-segment arrangement.

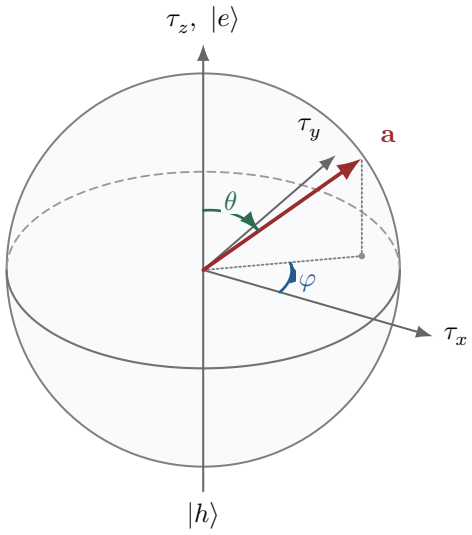


Figure 4.7: Nambu Bloch sphere in an orthographic projection. The example direction has $\theta = 55°$, $\varphi = 40°$; the dotted line projects it onto the equatorial plane. The polar angle is measured from $+\tau_z$, and the azimuth runs from $+\tau_x$ toward $+\tau_y$. The dashed half-ellipse is the rear equator. The poles denote electron and hole states; general components are given by equation (4.14).

For two Nambu analyzers let I_{ie}, I_{ih} denote positive-e flux contributions *after* the analyzer, so that $I_i^{\det} = I_{ie} - I_{ih}$. With the same time-domain and Fourier definitions as in equation (3.13), linearity gives

$$\begin{aligned} C_{12}^{\det}(u) &= C_{ee}(u) - C_{eh}(u) - C_{he}(u) + C_{hh}(u), \\ S_{12}^{\det}(\omega) &= S_{ee}(\omega) - S_{eh}(\omega) - S_{he}(\omega) + S_{hh}(\omega). \end{aligned} \tag{4.20}$$

The two minus signs come from opposite charge. Changing both detector signals into conventional electric currents changes neither cross spectrum. The C terms have units of current squared; the S terms have an additional unit of time and must not be equated to C.

For independent, well-characterized nonlocal pairs emitted at rate R, with one quasiparticle per side and complete detection,

$$S_{12}^{\det}(0) = e^2 R \sum_{s,t=e,h} z_s z_t p_{st}(\mathbf{a},\mathbf{b}) = e^2 R\, E(\mathbf{a},\mathbf{b}), \qquad z_e = +1,\ z_h = -1. \tag{4.21}$$

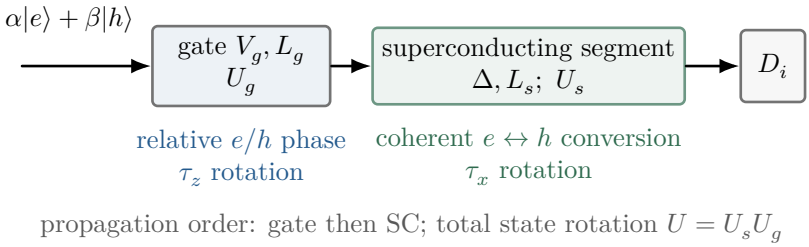


Figure 4.8: Ordered physical implementation of a Nambu analyzer. The gate changes the relative phase first; the superconducting conversion then makes that phase accessible to charge readout. Arrows denote propagation, not the sign of electric current. Electron and hole amplitudes are both retained; the superconducting segment is not assumed to convert perfectly.

This follows by integrating each pair's signed charge product and using Poisson event statistics. It is the same pair-rate normalization as in equation (3.32). A singlet gives $E = -\mathbf{a} \cdot \mathbf{b}$; for parallel axes the charge cross noise is negative, despite the positive spin-resolved pair-flux cross components of the preceding chapter.

Charge readout forms the signed sum in a single cross-noise measurement for each pair of settings; four separate charge-resolved detector outputs are unnecessary. It does not remove the need to determine R, to calibrate analyzer rotations, or to measure four setting pairs for a CHSH combination. Other emission statistics require the corresponding count/mean terms; local-pair and multipair backgrounds cannot simply be relabelled as the nonlocal pair rate. Chirality suppresses ideal backscattering but does not guarantee unit efficiency for real contacts, conversion or electronics.

Exercise 4.1. Why can a gate alone not serve as an arbitrary Nambu analyzer? *Answer:* $U_g^\dagger \tau_z U_g = \tau_z$, so it changes no charge probabilities. A noncommuting conversion, supplied here by the subsequent superconducting segment, is required to make the relative phase observable.

5 Fermionic beam splitters: bunching and antibunching

The purpose of this chapter is to connect exchange symmetry to detector records. A beam splitter converts symmetry of the incident two-particle state into output-number statistics. Turning those statistics into charge noise requires the charge of each excitation and a model of repeated injection. The spin example follows the mechanism discussed by Burkard, Loss, and Sukhorukov [12]; the electron–hole example illustrates the charge-sensitive inversion studied by Li and Chen [13].

5.1 Spinless input: the determinant enforces antibunching

Consider a two-input, two-output Hanbury Brown–Twiss (HBT) beam splitter with ordered ports $(1, 2)$ and $(3, 4)$,

$$\begin{pmatrix} b_3 \\ b_4 \end{pmatrix} = \mathbf{S} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \qquad \mathbf{S} = \begin{pmatrix} S_{31} & S_{32} \\ S_{41} & S_{42} \end{pmatrix}, \qquad \mathbf{S}^\dagger \mathbf{S} = \mathbb{1}. \tag{5.1}$$

The complex amplitudes, their ordered ports, and their phases are part of the definition; a "50:50" probability alone does not fix $\mathbf{S}$.

From $a^\dagger = b^\dagger \mathbf{S}$,

$$a_1^\dagger = S_{31} b_3^\dagger + S_{41} b_4^\dagger, \qquad a_2^\dagger = S_{32} b_3^\dagger + S_{42} b_4^\dagger. \tag{5.2}$$

For one spinless fermion in each input,

$$|\Psi_{\text{out}}\rangle = \left(S_{31}b_3^\dagger + S_{41}b_4^\dagger\right)\left(S_{32}b_3^\dagger + S_{42}b_4^\dagger\right)|0\rangle \tag{5.3}$$

$$= (S_{31}S_{42} - S_{32}S_{41})\, b_3^\dagger b_4^\dagger|0\rangle = (\det \mathbf{S}) b_3^\dagger b_4^\dagger|0\rangle. \tag{5.4}$$

Terms with two identical spinless fermions in one output vanish. Since $|\det \mathbf{S}| = 1$, one fermion exits each port: exchange produces perfect antibunching for this ideal two-particle input. Here each operator denotes one normalized incident or outgoing wave-packet mode; the two inputs have the same temporal and spectral envelope and no distinguishable internal label. The Pauli principle forbids two fermions in the *same mode*, not two fermions of different energy or arrival time in the same lead.

The physical subsystems used later to discuss spin entanglement are the two spatial input arms, with local spin rotations and spin-resolved detection in each arm. The spinless state $a_1^\dagger a_2^\dagger|0\rangle$ is a single Slater determinant: its HBT dip is exchange interference, not by itself an operational entanglement resource. The beam splitter subsequently mixes the two arms and is therefore not local relative to that input-arm partition.

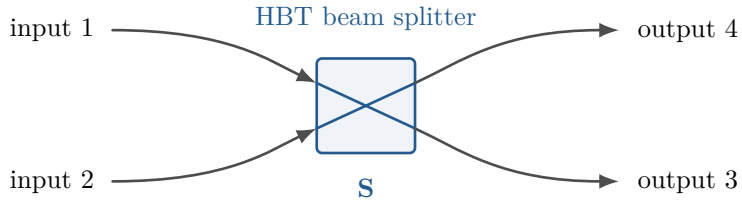


Figure 5.1: Port convention for the two-input, two-output HBT beam splitter. Arrows show propagation from input arms 1 and 2 to output arms 3 and 4; the matrix rows are ordered $(3, 4)$ even though output 4 is drawn above 3.

5.2 A spin singlet bunches at an ideal beam splitter

Take the spin singlet in distinct inputs,

$$|\Psi_{\text{in}}^-\rangle = \frac{a_{1\uparrow}^\dagger a_{2\downarrow}^\dagger - a_{1\downarrow}^\dagger a_{2\uparrow}^\dagger}{\sqrt{2}}|0\rangle. \tag{5.5}$$

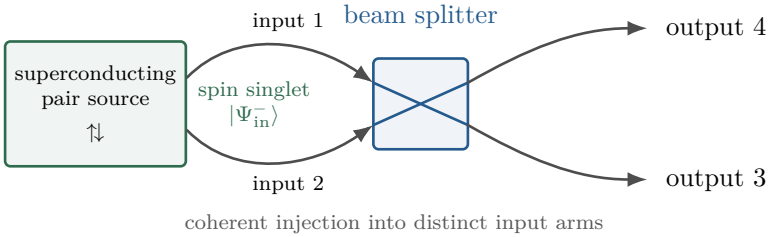


Figure 5.2: Ideal preparation of a split spin singlet from a superconducting pair source into inputs 1 and 2 of a common beam splitter. The diagram assumes successful split-pair emission; it does not assert unit source efficiency. The source ↑ / ↓ marker denotes the two spin components; their relative minus sign is fixed by equation (5.5).

Substitute equation (5.2) separately for each spin. Reordering creators into increasing output

port gives

$$|\Psi^{-}_{\text{out}}\rangle = \frac{1}{\sqrt{2}}[2S_{31}S_{32}b^{\dagger}_{3\uparrow}b^{\dagger}_{3\downarrow} + 2S_{41}S_{42}b^{\dagger}_{4\uparrow}b^{\dagger}_{4\downarrow} + (S_{31}S_{42} + S_{41}S_{32})\left(b^{\dagger}_{3\uparrow}b^{\dagger}_{4\downarrow} - b^{\dagger}_{3\downarrow}b^{\dagger}_{4\uparrow}\right)]|0\rangle. \tag{5.6}$$

For the real, balanced beam-splitter convention

$$\mathbf{S}_{50:50} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \tag{5.7}$$

the split-output amplitudes cancel and

$$|\Psi^{-}_{\text{out}}\rangle = \frac{b^{\dagger}_{3\uparrow}b^{\dagger}_{3\downarrow} - b^{\dagger}_{4\uparrow}b^{\dagger}_{4\downarrow}}{\sqrt{2}}|0\rangle. \tag{5.8}$$

The total fermionic state remains antisymmetric. Its spin-singlet factor is antisymmetric, so its orbital factor is symmetric and produces boson-like bunching. This factorized symmetry argument assumes spin-independent scattering and no unresolved degree of freedom that distinguishes the two incident wave packets.

For the $m = 0$ spin triplet, the spin factor is symmetric and the orbital factor is antisymmetric. Direct substitution gives

$$|\Psi^{+}_{\text{out}}\rangle = \frac{b^{\dagger}_{3\uparrow}b^{\dagger}_{4\downarrow} + b^{\dagger}_{3\downarrow}b^{\dagger}_{4\uparrow}}{\sqrt{2}}|0\rangle. \tag{5.9}$$

The pair remains split: it antibunches. Every ideal triplet event delivers exactly one particle to each output and therefore has no partition uncertainty. In contrast, a singlet delivers both electrons to output 3 or both to output 4 with equal probabilities. A selected-port pulse then carries charge $-2e$, with $e > 0$, but this does not change the charge of an individual electron or assert a universal Fano factor.

5.3 Electron–hole pairs invert the charge-noise intuition

For electron–hole pseudospin the two exchange symmetries may be written

$$|\Psi^{\mp}_{eh}\rangle = \frac{\gamma^{\dagger}_{1e}\gamma^{\dagger}_{2h} \mp \gamma^{\dagger}_{1h}\gamma^{\dagger}_{2e}}{\sqrt{2}}|0\rangle. \tag{5.10}$$

The upper minus sign is the Nambu singlet used in equation (4.10); the lower plus sign is the corresponding triplet-like combination.

For the following calculation, both excitation types obey the same real matrix in equation (5.7), with no electron–hole conversion at this beam splitter. The quasiparticle vacuum is the filled reference Fermi sea and all excitation energies are positive. We impose one canonical creation-operator order: increasing output port, and e before h within one port. Thus the same anticommutation algebra as in equations (5.6) and (5.9) gives

$$|\Psi^{-}_{eh,\text{out}}\rangle = \frac{\gamma^{\dagger}_{3e}\gamma^{\dagger}_{3h} - \gamma^{\dagger}_{4e}\gamma^{\dagger}_{4h}}{\sqrt{2}}|0\rangle, \qquad \text{bunched,} \tag{5.11}$$

$$|\Psi^{+}_{eh,\text{out}}\rangle = \frac{\gamma^{\dagger}_{3e}\gamma^{\dagger}_{4h} + \gamma^{\dagger}_{3h}\gamma^{\dagger}_{4e}}{\sqrt{2}}|0\rangle, \qquad \text{split.} \tag{5.12}$$

Number statistics call the first state bunched and the second antibunched. The relative signs above follow from the declared matrix and operator order; they cannot be changed by dropping

an overall phase. More generally, for a normal elastic scatterer in paired electron/hole bases, $\mathbf{S}_h(E) = \mathbf{S}_e^*(-E)$ follows by writing a hole annihilator as an electron creator below the Fermi level. A common real, energy-independent matrix satisfies this relation. Arbitrary scattering phases must instead be carried through explicitly; equal transmission probabilities alone do not determine which input Bell phase bunches.

Charge readout differs because electron and hole charges are $-e$ and $+e$. In a detector window containing the whole pair, the same-port electron–hole state has zero integrated charge in either output. The split state instead has two possible charge records, $(-e, +e)$ and $(+e, -e)$. Both have zero *mean* charge in each output, but only the split state has a nonzero charge variance. Mean current alone therefore does not distinguish them.

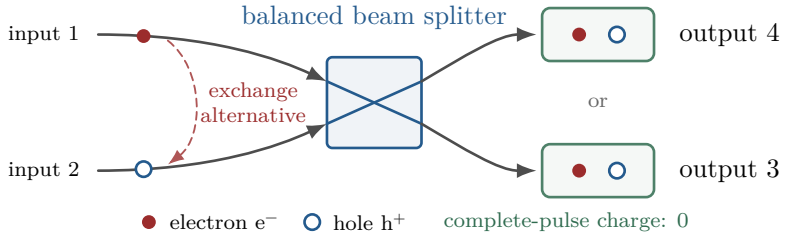


Figure 5.3: Nambu-singlet bunching at a balanced beam splitter. Filled and open markers denote an electron and a hole, and the dashed path indicates the two coherent exchange alternatives, not an additional particle trajectory. The outgoing neutral pair occupies either output 4 or output 3. Zero integrated charge does not imply zero time-resolved or finite-frequency fluctuations.

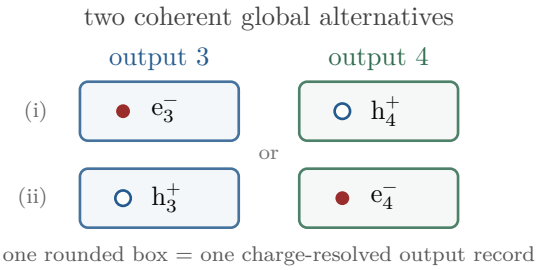


Figure 5.4: Nambu-triplet splitting into two detector outputs. Each row is one coherent two-output alternative: either an electron in output 3 and a hole in output 4, or a hole in output 3 and an electron in output 4. The rounded outlines group the charge-resolved pulses recorded at each output.

5.4 From a pair record to a noise measurement

Define the integrated output charge by $Q_j = \int_{\text{window}} I_j(t)\,\mathrm{d}t$, using the same outward physical-current direction at both outputs. For electrons $Q_j = -eN_j$; for electron–hole excitations $Q_j = e(N_{jh} - N_{je})$. With exactly one pair in each trial, the balanced splitter gives the following complete-window statistics:

For example, a spin singlet gives $Q_3 = -2e$ or 0, $\langle Q_3 \rangle = -e$, and $\operatorname{Var} Q_3 = 2e^2 - e^2 = e^2$. A Nambu triplet gives $Q_3 = \pm e$ and $Q_4 = -Q_3$, so its cross covariance is $-e^2$. Thus bunching enhances charge fluctuations for the spin example and suppresses them for the electron–hole example.

To connect this table to a spectrum, consider independent trials at rate f, with exactly one pair in each trial and the deterministic periodic mean removed. With the two-sided ordered

Input	(N_3, N_4) records	$\operatorname{Var} Q_3/e^2$	$\operatorname{Cov}(Q_3, Q_4)/e^2$
Spin singlet	$(2,0)$ or $(0,2)$	1	-1
Spin triplet	$(1,1)$	0	0
Nambu singlet	$(2,0)$ or $(0,2)$	0	0
Nambu triplet	$(1,1)$	1	-1

Table 5.1: Charge fluctuations for a fixed pair number and ideal balanced scattering. Alternatives have equal probabilities. N_j counts excitations irrespective of their charge; the last two columns describe charge, not excitation-number, fluctuations.

spectrum defined in equation (3.13), the cycle-averaged zero-frequency partition noise is

$$S_{ij}^{\mathrm{part}}(0) = f \operatorname{Cov}(Q_i, Q_j). \tag{5.13}$$

This follows because charge covariances add over independent trials, and the zero-frequency spectrum equals the long-time covariance per unit time. For a Poisson stream of pair events at rate r, source timing also fluctuates: instead, $S_{ij}(0) = r\langle Q_i Q_j\rangle$ for the event contribution. A spin triplet can then have nonzero source noise although its *partition* noise vanishes. Neither a noise sign nor an effective pulse charge can be interpreted without this distinction.

The zero-charge conclusion for a same-port electron–hole pair concerns a window containing both excitations, or the ideal zero-frequency pair contribution. If their pulse shapes or arrival times differ, instantaneous currents need not cancel and finite-frequency noise can survive. Real transport measurements additionally require equilibrium/background subtraction and the spectral convention introduced in section 3. The perfect exchange limits assume matched temporal and spectral modes, elastic energy-independent scattering, no loss, and no hidden which-input information. Imperfect overlap or electron–hole asymmetry invalidates the perfect limits.

Exercise 5.1 (Number noise versus charge noise)**.** For the Nambu singlet in equation (5.11), find $\operatorname{Var} N_3$ and compare it with $\operatorname{Var} Q_3$. *Answer:* $N_3 = 0$ or 2 with equal probabilities, so $\operatorname{Var} N_3 = 1$; every complete record has $Q_3 = 0$, so $\operatorname{Var} Q_3 = 0$. A number detector and a charge detector measure different observables.

6 Quantum erasure by correlated detection

Quantum erasure asks where interference can be found when a second system stores a path record. The answer is in suitably sorted joint records, while each unsorted local record obeys the no-signalling principle. This is the central idea of the quantum eraser [14]. A related mesoscopic proposal uses current correlations to probe spin entanglement with an interferometer and spin filter [15]; below we derive the ideal two-channel model represented in the source notes.

6.1 Reference: a one-particle interferometer

For a balanced two-path interferometer with relative phase φ, adopt the beam-splitter convention

$$\mathbf{S}_{\mathrm{BS}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}. \tag{6.1}$$

Here the first splitter prepares equal path amplitudes with $\mathbf{S}_{\mathrm{BS}}^{\dagger}$, the upper path acquires phase φ, and the recombiner uses $\mathbf{S}_{\mathrm{BS}}$. With columns ordered as upper and lower paths and output rows (D_1, D_2), the full transformation is

$$\begin{pmatrix} t_{D_1} \\ t_{D_2} \end{pmatrix} = \mathbf{S}_{\mathrm{BS}} \begin{pmatrix} \mathrm{e}^{\mathrm{i}\varphi} & 0 \\ 0 & 1 \end{pmatrix} \mathbf{S}_{\mathrm{BS}}^{\dagger} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 + \mathrm{e}^{\mathrm{i}\varphi} \\ 1 - \mathrm{e}^{\mathrm{i}\varphi} \end{pmatrix}. \tag{6.2}$$

Thus the amplitude at detector D_1 is

$$t_{D_1} = \frac{1}{2}\left(1 + e^{i\varphi}\right), \qquad P_{D_1} = |t_{D_1}|^2 = \frac{1}{2}(1 + \cos\varphi). \tag{6.3}$$

Both paths end in the same detected state, so their amplitudes interfere. The other output has $P_{D_2} = (1 - \cos\varphi)/2$ and the total probability is one. Choosing two identical matrices instead would exchange the bright and dark ports at $\varphi = 0$; port phases must be specified along with a "balanced" splitter.

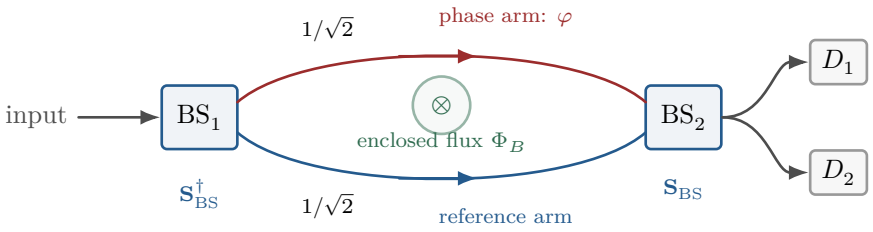


Figure 6.1: Balanced one-particle interferometer. The first beam splitter creates two path amplitudes, one acquires the relative phase φ, and the second beam splitter recombines them at D_1. The $\otimes$ marks flux into the page. Its signed phase contribution is $q\Phi_B/\hbar$, where the orientation defines Φ_B and $q = -e$ for an electron; any dynamical path phase is included in φ.

6.2 Which-way information removes the marginal fringe

The emitted two-particle state is normalized as

$$|\Psi_{\mathrm{src}}\rangle = \sqrt{\eta}|\uparrow\rangle_1|\downarrow\rangle_2 - \sqrt{1-\eta}|\downarrow\rangle_1|\uparrow\rangle_2, \qquad 0 \le \eta \le 1. \tag{6.4}$$

The subsystem labels 1 and 2 refer to two spatial regions with one electron each; the tensor-product order is region 1 then region 2. Here η and $1-\eta$ are the probabilities of the two emission alternatives, whereas $\sqrt{\eta}$ and $\sqrt{1-\eta}$ are their probability amplitudes. The ideal equal-weight singlet corresponds to $\eta = 1/2$; other values describe unequal prepared emission amplitudes. In the ideal helical-edge interpretation, opposite members of a Kramers pair propagate in opposite directions on each edge. The arrows label an ideal conserved spin axis as well as the corresponding channels. In a generic spin–orbit-coupled edge they are channel labels, not necessarily pure eigenstates of physical σ_z.

In the two-region geometry, use detector labels for the output basis. The local phase plus recombiner in region 1 acts as

$$|\uparrow\rangle_1 \longmapsto \frac{|D_1\rangle_1 + |D_3\rangle_1}{\sqrt{2}}, \qquad |\downarrow\rangle_1 \longmapsto \frac{-|D_1\rangle_1 + |D_3\rangle_1}{\sqrt{2}}e^{i\varphi}. \tag{6.5}$$

Its matrix is $U_1 = \mathbf{S}^\dagger_{\mathrm{BS}}\,\mathrm{diag}(1, e^{i\varphi})$ in input order $(\uparrow_1, \downarrow_1)$ and output order (D_1, D_3). This local splitter has its own ordered ports; it is not the entire two-splitter network in equation (6.2). Substitution gives the expanded two-particle state

$$\begin{aligned}|\Psi_1\rangle = &\sqrt{\frac{\eta}{2}}|D_1\rangle_1|\downarrow\rangle_2 + \sqrt{\frac{1-\eta}{2}}e^{i\varphi}|D_1\rangle_1|\uparrow\rangle_2 \\ &+ \sqrt{\frac{\eta}{2}}|D_3\rangle_1|\downarrow\rangle_2 - \sqrt{\frac{1-\eta}{2}}e^{i\varphi}|D_3\rangle_1|\uparrow\rangle_2.\end{aligned} \tag{6.6}$$

The first two terms reach D_1; their particle-2 states are orthogonal and therefore carry perfect which-way information.

Tracing out particle 2 gives

$$P_{D_1} = \langle D_1|_1 \,\mathrm{Tr}_2 |\Psi_1\rangle\langle\Psi_1||D_1\rangle_1 \tag{6.7}$$

$$= \frac{\eta}{2} + \frac{1-\eta}{2} + \sqrt{\eta(1-\eta)}\,\mathrm{Re}[e^{i\varphi}\langle\downarrow_2|\uparrow_2\rangle]$$

$$= \frac{\eta}{2} + \frac{1-\eta}{2} = \frac{1}{2}. \tag{6.8}$$

The marginal current at D_1 has no phase dependence. This is loss of single-particle coherence after an unobserved, orthogonal which-way marker is traced out; it does not require random dynamical fluctuations of φ.

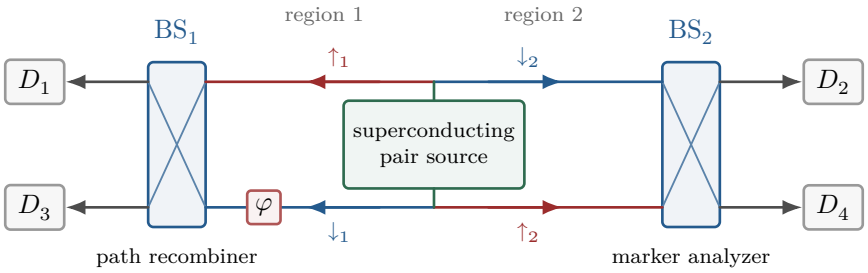


Figure 6.2: Ideal two-region quantum eraser in a helical-channel interpretation. The central superconducting pair source emits the two spin–path alternatives in equation (6.4); the selected channels send particle 1 left and particle 2 right. Beam splitter 1 recombines the phase-tagged paths, while beam splitter 2 selects a complementary marker basis. The D_1 marginal is phase independent, whereas beam-splitter-2-sorted coincidences display complementary fringes. Output rows are (D_1, D_3) on the left and (D_2, D_4) on the right; output port names do not specify the spin entering the analyzer. The diagram assumes coherent channel mixing and successful pair splitting; it is not a complete device design.

6.3 The eraser basis recovers conditional fringes

To erase particle 2's which-way record, BS_2 measures it in the rotated basis

$$|\uparrow\rangle_2 \longmapsto \frac{|D_2\rangle_2 + |D_4\rangle_2}{\sqrt{2}}, \qquad |\downarrow\rangle_2 \longmapsto \frac{-|D_2\rangle_2 + |D_4\rangle_2}{\sqrt{2}}. \tag{6.9}$$

In input order $(\uparrow_2, \downarrow_2)$ and output order (D_2, D_4), the analyzer matrix is $U_2 = \mathbf{S}_{\mathrm{BS}}^{\dagger}$. Detecting D_2 or D_4 is thus a projection onto the incoming marker states $(|\uparrow\rangle_2 - |\downarrow\rangle_2)/\sqrt{2}$ or $(|\uparrow\rangle_2 + |\downarrow\rangle_2)/\sqrt{2}$, respectively. Keeping the D_1 part of equation (6.6) and applying equation (6.9) gives

$$|\Psi_{D_1}\rangle = \frac{\sqrt{\eta}}{2}|D_1\rangle_1 \left(-|D_2\rangle_2 + |D_4\rangle_2\right)$$

$$+ \frac{\sqrt{1-\eta}}{2} e^{i\varphi} |D_1\rangle_1 \left(|D_2\rangle_2 + |D_4\rangle_2\right). \tag{6.10}$$

This is an unnormalized D_1 component of norm squared $1/2$. The two joint amplitudes are therefore

$$\alpha_{12} = -\frac{\sqrt{\eta}}{2} + \frac{\sqrt{1-\eta}}{2} e^{i\varphi}, \tag{6.11}$$

$$\alpha_{14} = \frac{\sqrt{\eta}}{2} + \frac{\sqrt{1-\eta}}{2} e^{i\varphi}. \tag{6.12}$$

Their probabilities are

$$P(D_1, D_2) = |\alpha_{12}|^2 = \frac{1}{4}\left[1 - 2\sqrt{\eta(1-\eta)}\cos\varphi\right], \tag{6.13}$$

$$P(D_1, D_4) = |\alpha_{14}|^2 = \frac{1}{4}\left[1 + 2\sqrt{\eta(1-\eta)}\cos\varphi\right], \tag{6.14}$$

$$P(D_1, D_2) + P(D_1, D_4) = \frac{1}{2}. \tag{6.15}$$

For the normalized pure two-qubit state in equation (6.4), the coefficient

$$C = 2\sqrt{\eta(1-\eta)} \tag{6.16}$$

is its concurrence.

The complete joint probability distribution, with rows (D_1, D_3) and columns (D_2, D_4), is

$$\begin{pmatrix} P(D_1, D_2) & P(D_1, D_4) \\ P(D_3, D_2) & P(D_3, D_4) \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 1 - C\cos\varphi & 1 + C\cos\varphi \\ 1 + C\cos\varphi & 1 - C\cos\varphi \end{pmatrix}. \tag{6.17}$$

Every row and column sums to $1/2$. Thus all four detector marginals are phase independent, including $P(D_2) = P(D_4) = 1/2$. The normalized *conditional* probabilities are

$$P(D_1 \mid D_2) = \frac{P(D_1, D_2)}{P(D_2)} = \frac{1}{2}(1 - C\cos\varphi), \tag{6.18}$$

$$P(D_1 \mid D_4) = \frac{P(D_1, D_4)}{P(D_4)} = \frac{1}{2}(1 + C\cos\varphi). \tag{6.19}$$

The factor of two is essential: a coincidence probability counts a fraction of *all* emitted pairs, whereas a conditional probability counts a fraction of pairs with the selected marker result.

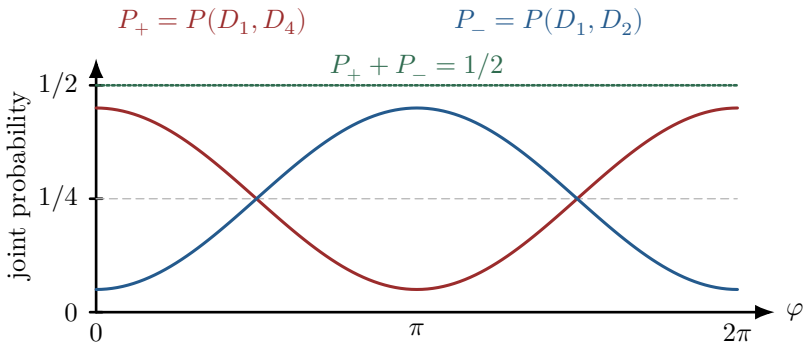


Figure 6.3: Quantitative complementary coincidence fringes for $\eta = 1/5$, hence $C = 4/5$. Curves are computed from equation (6.17): their mean is $1/4$, their amplitude is $C/4 = 1/5$, and their extrema are $1/20$ and $9/20$. Their sum is the flat D_1 marginal. Conditional probabilities are twice the plotted joint probabilities.

6.4 When visibility measures concurrence

Writing either coincidence fringe as

$$P_\pm(\varphi) = \frac{1}{4}\left[1 \pm V_{\text{vis}}\cos\varphi\right], \qquad V_{\text{vis}} \equiv \frac{P_{\max} - P_{\min}}{P_{\max} + P_{\min}}, \tag{6.20}$$

where $P_+ = P(D_1, D_4)$ and $P_- = P(D_1, D_2)$, the two probabilities above give $V_{\text{vis}} = C$ in the ideal setup. Conditional fringes have the same visibility here because their normalization is phase

independent. This equality assumes the pure input in equation (6.4), orthogonal marker states, the balanced, lossless transformations equations (6.5) and (6.9), a stable relative phase, ideal projective detection, and no additional loss or dephasing. For a mixed input state, unbalanced beam splitters, unequal path transmission, or imperfect markers, the measured visibility need not equal concurrence.

The factor $2\sqrt{\eta(1-\eta)}$ is twice the product of the two emission amplitudes. It vanishes at $\eta = 0$ or 1, where only one alternative is present, and reaches unity at $\eta = 1/2$, where the alternatives have equal weight. This is why the same factor controls both pure-state entanglement and the contrast of the ideally sorted interference fringes.

An unbalanced marker projection gives a useful counterexample to a general $V_{\mathrm{vis}} = C$ rule. Replace the complementary marker result by $|m\rangle = \cos\theta|{\downarrow}\rangle_2 + \sin\theta|{\uparrow}\rangle_2$, with $0 \le \theta \le \pi/2$. Its probability and its sorted-fringe visibility are

$$p_m = \eta\cos^2\theta + (1-\eta)\sin^2\theta, \tag{6.21}$$

$$V_{\mathrm{vis},m} = \frac{2\sqrt{\eta(1-\eta)}\cos\theta\sin\theta}{p_m}, \qquad p_m > 0. \tag{6.22}$$

The numerator is twice the product of the conditional path amplitudes; the denominator is their total weight. At $\theta = \pi/4$ this gives $V_{\mathrm{vis},m} = C$, but choosing $\tan\theta = \sqrt{\eta/(1-\eta)}$ for $0 < \eta < 1$ makes the two conditional path amplitudes equal and gives $V_{\mathrm{vis},m} = 1$, even when $C < 1$. The price is the selection probability $p_m = 2\eta(1-\eta)$. A high-contrast selected fringe is therefore not a general entanglement measure or, by itself, a Bell test.

6.5 Coincidences, connected noise, and no signalling

For one detected pair per trial, let n_i be the indicator of a count at D_i. The distinction from connected correlations is explicit:

$$\langle n_1 n_2\rangle = P(D_1, D_2), \qquad \langle \delta n_1 \delta n_2\rangle = P(D_1, D_2) - \frac{1}{4} = -\frac{C}{4}\cos\varphi. \tag{6.23}$$

The covariance has no positive coincidence offset. For independent, fixed-rate pair trials its charge-noise contribution is fe^2 times this covariance. For Poisson pair emission at rate r, instead, the event contribution is $S_{12}(0) = re^2 P(D_1, D_2)$ under the same outward-current and spectral conventions as equation (5.13). Thus a calibrated noise measurement can reveal the joint fringe, but noise is not automatically a probability or a normalized conditional record. Accidental coincidences, loss, source statistics, and background subtraction must be included before extracting a visibility from transport data.

Quantum erasure does not retroactively change an already recorded marginal event, signal from particle 2 to particle 1, or restore interference in P_{D_1}. It changes which joint observable is inspected: discarding the which-way outcome gives equation (6.8), whereas conditioning on a complementary particle-2 basis gives equation (6.19). Sorting requires the two regions to compare their records by ordinary classical communication; changing a local analyzer cannot alter the other region's marginal distribution.

Exercise 6.1 (A normalized eraser record)**.** For $\eta = 1/5$ and $\varphi = 0$, find $P(D_1, D_2)$, $P(D_1 \mid D_2)$, and $P(D_1)$. *Answer:* $C = 4/5$, so the probabilities are 1/20, 1/10, and 1/2, respectively. The first two vary with phase and the third does not. At $\eta = 0$ or 1, $C = 0$ and all four joint probabilities are 1/4 for these balanced analyzers.

Summary. These notes move from a declared bipartition and Schmidt spectrum to a Bell witness, then to mesoscopic source-and-analyzer design and current-noise readout. Orbital and Nambu pseudospins, beam-splitter exchange, and quantum erasure reuse this logic only after their physical carriers and detector observables have been fixed.

A Notation and selected references

A.1 Notation

Symbol	Meaning and domain
$\mathrm{H}_{AB} = \mathrm{H}_A \otimes \mathrm{H}_B$	Composite Hilbert space and its declared subsystem factors.
$\rho_{AB}, \rho_A, \rho_B$	Joint state and reduced states.
s_i	Singular values of the pure-state coefficient matrix.
$\lambda_i = s_i^2$	Schmidt probabilities; nonzero entries are positive, $\sum_i \lambda_i = 1$; zero entries may be appended.
S_A	von Neumann entropy of ρ_A; natural logarithm unless bits are stated explicitly.
C	Concurrence of a two-qubit state.
$E(\mathbf{a}, \mathbf{b})$	Binary-outcome correlator, not a probability.
$\mathbf{a}, \mathbf{b}$	Bold italic unit analyzer directions, $\|\mathbf{a}\| = \|\mathbf{b}\| = 1$.
$\beta(\lambda), B_{\mathrm{CHSH}}$	Pointwise and ensemble-averaged CHSH combinations, respectively.
$\boldsymbol{\sigma}$	Pauli vector acting on physical spin.
$\boldsymbol{\mu}$	Pauli vector acting on an orbital or lead two-level system.
$\boldsymbol{\tau}$	Pauli vector in the ordered Nambu basis (e, h), with $q_{\mathrm{N}} = -e\tau_z$.
$\mathbf{S}$	Ordered-port scattering matrix of complex amplitudes.
$\mathbf{N}, \mathbf{I}$	Counting and current operators, written bold upright without hat accents.
$C_{\alpha\beta}(u), S_{\alpha\beta}(\omega)$	Connected time correlation (A^2) and two-sided ordered spectrum ($\mathrm{A}^2\,\mathrm{s}$), as defined in equation (3.13).
S_0	$2\|\Gamma\|^2 e^3 V/h$ in the weak-pairing model, with no extra factor of two in the spectrum.
Γ, R	Dimensionless weak-pairing amplitude and pair-emission rate in Section 3. The positive matrix R in Section 1 is a separate, locally defined concurrence construction.
$q, p_{\alpha\beta}$	Pair probability per gate and conditional outcome probability given a pair, respectively.
$T, \mathcal{T}$	Counting gate and observation time for energy-bin regularization; $\delta E = h/\mathcal{T}$.
η	Weight of one quantum-eraser emission alternative, $0 \le \eta \le 1$.
$V, V(t), V_g$	Bias voltage, time-dependent drive, and gate voltage, respectively.
V_{vis}	Visibility of a conditional interference fringe.

Table A.1: Notation used throughout the notes.

Reading the references. The references cited in the chapters support the specified definitions or physical proposals. For the quantum-dot spin-qubit motivation, see also Ref. [16]. The models in Sections 3–6 remain subject to the detector, phase, and state-selection assumptions stated there.

References

[1] A. Einstein, B. Podolsky, and N. Rosen, "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?," *Phys. Rev.* **47**, 777–780 (1935). doi:10.1103/PhysRev.47.777.

[2] J. S. Bell, "On the Einstein Podolsky Rosen paradox," *Physics Physique Fizika* **1**, 195–200 (1964). doi:10.1103/PhysicsPhysiqueFizika.1.195.

[3] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, "Proposed Experiment to Test Local Hidden-Variable Theories," *Phys. Rev. Lett.* **23**, 880–884 (1969). doi:10.1103/PhysRevLett.23.880.

[4] W. K. Wootters, "Entanglement of Formation of an Arbitrary State of Two Qubits," *Phys. Rev. Lett.* **80**, 2245–2248 (1998). doi:10.1103/PhysRevLett.80.2245.

[5] H. Ollivier and W. H. Zurek, "Quantum Discord: A Measure of the Quantumness of Correlations," *Phys. Rev. Lett.* **88**, 017901 (2002). doi:10.1103/PhysRevLett.88.017901.

[6] R. F. Werner, "Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model," *Phys. Rev. A* **40**, 4277–4281 (1989). doi:10.1103/PhysRevA.40.4277.

[7] H. M. Wiseman and J. A. Vaccaro, "Entanglement of Indistinguishable Particles Shared between Two Parties," *Phys. Rev. Lett.* **91**, 097902 (2003). doi:10.1103/PhysRevLett.91.097902.

[8] P. Recher, E. V. Sukhorukov, and D. Loss, "Andreev tunneling, Coulomb blockade, and resonant transport of nonlocal spin-entangled electrons," *Phys. Rev. B* **63**, 165314 (2001). doi:10.1103/PhysRevB.63.165314.

[9] N. M. Chtchelkatchev, G. Blatter, G. B. Lesovik, and T. Martin, "Bell inequalities and entanglement in solid-state devices," *Phys. Rev. B* **66**, 161320(R) (2002). doi:10.1103/PhysRevB.66.161320.

[10] C. W. J. Beenakker, C. Emary, M. Kindermann, and J. L. van Velsen, "Proposal for Production and Detection of Entangled Electron-Hole Pairs in a Degenerate Electron Gas," *Phys. Rev. Lett.* **91**, 147901 (2003). doi:10.1103/PhysRevLett.91.147901.

[11] W. Luo, H. Geng, D. Y. Xing, G. Blatter, and W. Chen, "Entanglement of Nambu Spinors and Bell Inequality Test without Beam Splitters," *Phys. Rev. Lett.* **129**, 120507 (2022). doi:10.1103/PhysRevLett.129.120507.

[12] G. Burkard, D. Loss, and E. V. Sukhorukov, "Noise of entangled electrons: Bunching and antibunching," *Phys. Rev. B* **61**, R16303–R16306 (2000). doi:10.1103/PhysRevB.61.R16303.

[13] H.-D. Li and W. Chen, "Anomalous exchange correlation of quasiparticles with entangled Nambu spinors," *Phys. Rev. B* **111**, 195306 (2025). doi:10.1103/PhysRevB.111.195306.

[14] M. O. Scully and K. Drühl, "Quantum eraser: A proposed photon correlation experiment concerning observation and 'delayed choice' in quantum mechanics," *Phys. Rev. A* **25**, 2208–2213 (1982). doi:10.1103/PhysRevA.25.2208.

[15] W. Chen, R. Shen, Z. D. Wang, L. Sheng, B. G. Wang, and D. Y. Xing, "Quantitatively probing two-electron entanglement with a spintronic quantum eraser," *Phys. Rev. B* **87**, 155308 (2013). doi:10.1103/PhysRevB.87.155308.

[16] D. Loss and D. P. DiVincenzo, "Quantum computation with quantum dots," *Phys. Rev. A* **57**, 120–126 (1998). doi:10.1103/PhysRevA.57.120.